Hamilton cycles at the minimum-degree-two hitting time

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October 18, 2026

Abstract

The random graph process starts with an empty graph on n vertices and adds a uniformly chosen missing edge at each step. We give an entropy proof of the classical theorem that, with high probability, a Hamilton cycle appears when the minimum degree first reaches 2. With high probability, the number of Hamilton cycles at this time τ_2 is $(2\tau_2/(en))^n \exp\{O(n/\log n)\}$. The proof adapts conditioned sampling and random edge-deletion arguments from Kahn's work on perfect matchings. An entropy inequality for random perfect matchings in bipartite graphs and injective cycle modifications give a uniform bound on the fraction of Hamilton cycles containing any edge.

1 Introduction

A *Hamilton cycle* in a graph is a cycle containing every vertex. Let $(G_t)_{t=0}^{\binom{n}{2}}$ be the random graph process on $[n] \coloneqq \{1, \dots, n\}$. More precisely, set $G_0 = \varnothing$ and, conditional on G_t, add one edge chosen uniformly from $\binom{[n]}{2} \setminus G_t$ to obtain G_{t+1}. We define two hitting times in this process: τ_2 is the first time the minimum degree reaches 2, and τ_{HC} is the first time a Hamilton cycle appears. That is,

$$\tau_2 \coloneqq \min\{t \colon \delta(G_t) \geq 2\}, \quad \text{and} \quad \tau_{\mathrm{HC}} \coloneqq \min\{t \colon G_t \text{ contains a Hamilton cycle}\}.$$

Observe that every vertex of a Hamilton cycle has degree 2, so $\tau_{\mathrm{HC}} \geq \tau_2$ deterministically. Bollobás [3] and, independently, Ajtai, Komlós, and Szemerédi [1] proved that these hitting times coincide with high probability. We give a proof using entropy, conditional sampling, and modifications of Hamilton cycles.

Theorem 1.1. *As n tends to infinity, $\mathbb{P}[\tau_{\mathrm{HC}} = \tau_2]$ tends to 1.*

For a graph G, let $Z(G)$ denote its number of Hamilton cycles. The proof also estimates this number at the hitting time, retaining the actual value of τ_2 in the main term.

Theorem 1.2. *There is an absolute constant $C > 0$ such that, with probability $1 - o(1)$,*

$$\left| \log Z(G_{\tau_2}) - \left(n \log \frac{2\tau_2}{n} - n \right) \right| \leq \frac{Cn}{\log n}. \tag{1.1}$$

In particular, with high probability, $Z(G_{\tau_2}) = (\log n/e)^n \exp\{o(n)\}$.

Glebov and Krivelevich [6, Theorem 2] proved the leading exponential estimate $(\log n/e)^n \exp\{o(n)\}$ at the minimum-degree-two hitting time. The estimate in (1.1) gives an error of order $n/\log n$ for the logarithm of the number of Hamilton cycles, with the reference value expressed in terms of the actual hitting time.

The proof adapts entropy, conditioned sampling, and random edge-deletion arguments used by Kahn [11, 10] for perfect matchings. We first reveal all edges incident to vertices of very small degree at τ_2. For each such vertex v, choose two neighbors a, b and replace the path $a - v - b$ by a prescribed pair ab. These paths are vertex-disjoint with high probability. Conditional on the revealed edges, the graph induced by the remaining vertices is uniform subject to its number of edges and the resulting degree lower bounds. A Hamilton cycle through all prescribed pairs expands to a Hamilton cycle in G_{τ_2}.

We count cycles through prescribed pairs by randomly deleting edges from the complete graph down to this conditioned graph. The main task is to bound the fraction of counted cycles that contain any one edge. An entropy inequality for random perfect matchings in bipartite graphs first shows that most directed edge marginals are nearly equal. Comparing the forward and reverse laws of a random edge-deletion experiment extends this estimate to prescribed pairs that need not be edges of the host graph. Injective

cycle modifications then compare the numbers of cycles through different prescribed pairs and yield a bound for every edge marginal. A martingale estimate controls the accumulated decrease in the logarithm of the number of cycles.

The inequality for perfect matchings in bipartite graphs follows the random-order entropy argument of Radhakrishnan [12]. Cuckler and Kahn [5] developed entropy bounds for perfect matchings and Hamiltonian cycles in graphs using edge marginals. Here the entropy estimate is applied to cycles through prescribed pairs in a random graph subject to degree lower bounds. The modifications include the reversal of a path segment, but the proof does not use Pósa's rotation–extension method to extend a path.

Sections 3 and 4 establish the degree-conditioned model and the reduction at τ_2. The entropy estimates, conditional sampling laws, and cycle modifications in Sections 5 to 8 give a uniform edge-marginal bound. We complete the counting proofs in Section 9.

2 Preliminaries

We collect the graph notation, the enumeration of cycles through prescribed pairs, and the probability and entropy tools used below.

All graphs are simple and undirected. We usually identify a graph G with its edge set and write $|G|$ for its number of edges. For $S \subseteq V(G)$, let $G - S$ be the graph induced by $V(G) \setminus S$, and write $G - v$ for $G - \{v\}$. For an edge set $T \subseteq G$, write $G - T$ for the graph obtained by deleting the edges of T, with its vertex set unchanged; for a single edge e, write $G - e$ for $G - \{e\}$.

The neighborhood of v is $N_G(v) \coloneqq \{u \in V(G) \setminus \{v\} \colon uv \in G\}$, and its degree is $d_G(v) \coloneqq |N_G(v)|$. Write $\delta(G)$ and $\Delta(G)$ for the minimum and maximum degrees, and $d(G) \coloneqq 2|G|/|V(G)|$ for the average degree.

We need Hamilton cycles containing prescribed pairs, including pairs absent from the host graph.

Definition 2.1. Let G be a graph on a vertex set V, and let M be a prescribed matching on V.

(i) A (G, M)-*respecting Hamilton cycle* is a Hamilton cycle on V containing every pair in M, with all its other edges in $G \setminus M$.

(ii) A vertex in $V(M)$ is M-*saturated*; all other vertices are M-*unsaturated*.

(iii) Write $\Phi(G, M)$ for the number of these cycles. For such a cycle C, the edges in $C \setminus M$ are its *ordinary edges*.

Thus each counted cycle uses $|V| - |M|$ ordinary edges. When $M \neq \varnothing$, choose a *root pair* and fix an arbitrary direction on it. Each counted cycle has a unique orientation agreeing with this root direction.

For a fixed directed root pair in M and distinct $u, v \in V(G) \setminus V(M)$, let $\Psi_{G,M}(u \to v)$ count the $(G, M \cup \{uv\})$-respecting Hamilton cycles in which the new pair is directed from u to v relative to the root direction. This number is defined whether or not $uv \in G$.

The following lemma counts Hamilton cycles through a prescribed matching when the host graph is complete.

Lemma 2.2. *Let M be a matching of $s \geq 1$ pairs on an n-vertex set V, with $n - s \geq 3$. Then*

$$\Phi\left(\binom{V}{2}, M\right) = 2^{s-1}(n-s-1)!.$$

If $n \geq 3s$, then requiring every ordinary edge to contain at most one M-saturated vertex multiplies this number by

$$\frac{(n-2s)!(n-2s-1)!}{(n-s-1)!(n-3s)!} \geq 1 - \frac{s(s-1)}{n-s-1}.$$

In particular, if $s = o(\sqrt{n})$, excluding ordinary edges between M-saturated vertices preserves a $1 - o(1)$ fraction of these cycles.

Proof. Contract each prescribed pair to a block, and regard every unsaturated vertex as a singleton block. There are $n - s$ blocks. Fixing the root direction gives $(n - s - 1)!$ cyclic orders of the blocks and 2^{s-1} choices of direction for the other prescribed pairs. Expanding the blocks determines the cycle uniquely, proving the first assertion.

The restriction in the second assertion requires that no two prescribed-pair blocks be consecutive. There are $n-2s$ singleton blocks. Arrange them in a cyclic order, choose s of the positions between consecutive singleton blocks, and insert the s distinct prescribed-pair blocks in those positions. This gives $(n-2s-1)!\binom{n-2s}{s}s!$ block arrangements, whereas the unrestricted number is $(n-s-1)!$. Their ratio is the claimed fraction. Each prescribed-pair block has a prescribed-pair block as its successor with probability $(s-1)/(n-s-1)$. A union bound over the s blocks proves the lower bound. □

When $G = \binom{V}{2}$, $|V| = n$, $|M| = s \geq 1$, and $n - s \geq 4$, the same enumeration gives $\Psi_{G,M}(u \to v)/\Phi(G, M) = 1/(n-s-1)$ for distinct $u, v \in V \setminus V(M)$. Indeed, reversing the added pair block gives a bijection between its two relative orientation classes, each of size $2^{s-1}(n-s-2)!$. If $s = o(n)$, this ratio is $(1 + o(1))/d(G)$. Under an entropy lower bound, Theorem 5.1 will give the corresponding estimate for most oriented edges of a sparse host.

Fact 2.3. *Let M be nonempty and fix a directed root pair as above. For distinct $u, v \in V(G) \setminus V(M)$, the following statements hold.*

(i) We have

$$\Phi(G, M \cup \{uv\}) = \Psi_{G,M}(u \to v) + \Psi_{G,M}(v \to u).$$

(ii) If $uv \in G$, then the number of (G, M)-respecting Hamilton cycles containing uv is $\Phi(G, M \cup \{uv\})$.

The first identity partitions the cycles according to the relative direction of uv. The second designates the ordinary edge uv as a prescribed pair without changing the underlying cycle.

We now introduce the probability notation. An event holds *with high probability* (whp) if its probability tends to 1 as the number of vertices tends to infinity. For real random variables X, Y, write $X \preceq Y$ if $\mathbb{P}[X \geq t] \leq \mathbb{P}[Y \geq t]$ for every real t; in other words, Y stochastically dominates X. All logarithms are natural, and we use $\log 0 = -\infty$ in inequalities involving cycle numbers.

For a nonempty finite set S, write $\mathrm{Unif}(S)$ for its uniform distribution, and write $\mathrm{Poisson}(\lambda)$ for the Poisson distribution of mean λ. If T is a uniformly chosen t-element subset of a finite set U and $A \subseteq U$ is fixed, then $|T \cap A|$ has a *hypergeometric distribution*.

A value z of a finite-valued random vector Z is $\mathbb{P}$-*admissible* if $\mathbb{P}[Z = z] > 0$. Whenever we condition on values of finite-valued random variables, those values are assumed to be admissible for the indicated law.

The following standard tail bounds are given, for example, by Janson, Łuczak, and Ruciński [9, Chapter 2]. Hypergeometric variables satisfy the same exponential-moment bounds as sampling with replacement by Hoeffding's comparison [8, Theorem 4].

Lemma 2.4. *Let X be binomial, hypergeometric, or Poisson with mean $\lambda > 0$, and put $\varphi(a) := 1 - a + a\log a$ for $a > 0$. The following statements hold.*

(i) For $0 < a < 1$, $\mathbb{P}[X \leq a\lambda] \leq \exp\{-\lambda\varphi(a)\}$.

(ii) For $t \geq \lambda$, $\mathbb{P}[X \geq t] \leq \exp\{-t\log(t/\lambda) + t - \lambda\} \leq (e\lambda/t)^t$.

(iii) For $0 < \varepsilon \leq 1$, $\mathbb{P}[|X - \lambda| \geq \varepsilon\lambda] \leq 2\exp\{-\varepsilon^2\lambda/3\}$.

For a nonnegative integer-valued $X \preceq \mathrm{Poisson}(\lambda)$, we also use $\mathbb{P}[X \geq t] \leq \lambda^t/t!$ for every positive integer t. Indeed, for $Y \sim \mathrm{Poisson}(\lambda)$, Markov's inequality applied to $\binom{Y}{t}$ gives $\mathbb{P}[Y \geq t] \leq \mathbb{E}[\binom{Y}{t}] = \lambda^t/t!$. Under independent Bernoulli sampling of edges, increasing events are positively associated by Harris's inequality [7].

For a probability vector $p = (p_z)_z$ on a finite set, define its entropy by $\mathbb{H}[p] := -\sum_z p_z \log p_z$. For a finite-valued random variable Z, we write $\mathbb{H}[Z]$ for the entropy of its distribution. For probability distributions p, q on the same finite set, define

$$D_{\mathrm{KL}}[p\|q] := \sum_z p_z \log \frac{p_z}{q_z},$$

with $0 \log 0 = 0$ and a term with $p_z > 0 = q_z$ interpreted as $+\infty$. Conditional entropy is the average of the entropies under the conditional laws. We use the chain rule $\mathbb{H}[Z, W] = \mathbb{H}[W] + \mathbb{H}[Z \mid W]$, subadditivity of entropy, nonnegativity of relative entropy, and the log-sum inequality; see Cover and Thomas [4, Chapter 2].

Lemma 2.5 (Pinsker's inequality)**.** *For probability distributions p, q on a finite set, $\sum_z |p_z - q_z| \le \sqrt{2 D_{\mathrm{KL}}[p\|q]}$.*

3 Random graphs with degree lower bounds

A key object in the proof is the family of graphs satisfying prescribed degree lower bounds. For a positive integer n, a nonnegative integer m, and a vector $\boldsymbol{\ell} = (\ell_v)_{v\in[n]}$ of nonnegative integers, define

$$\Omega(n, m, \boldsymbol{\ell}) \coloneqq \left\{ G \subseteq \binom{[n]}{2} : |G| = m \text{ and } d_G(v) \ge \ell_v \text{ for every } v \in [n] \right\}.$$

We call $(n, m, \boldsymbol{\ell})$

- *admissible* if $\Omega(n, m, \boldsymbol{\ell}) \neq \varnothing$;
- *near-critical*, with a fixed offset bound $A \ge 0$, if

$$\left| m - \frac{n \log n}{2} \right| \le \frac{3n \log\log n}{2}, \quad \text{and} \quad |\ell_v - \lfloor 0.01 \log n \rfloor| \le A \quad \text{for every } v \in [n].$$

All asymptotic statements in this section are uniform over these parameters for fixed A.

We first use a switching to bound the degree of a specified vertex.

Lemma 3.1. *Let G be uniform in a nonempty $\Omega(n, m, \boldsymbol{\ell})$. For every $v \in [n]$,*

$$\max\left\{0, d_G(v) - \max\left\{\ell_v, \left\lceil \frac{2m}{n} \right\rceil\right\}\right\} \preceq \mathrm{Poisson}\left(\frac{2m}{n}\right).$$

Consequently, for every integer $t \ge 2m/n > 0$,

$$\mathbb{P}\left[d_G(v) \ge \max\left\{\ell_v, \left\lceil \frac{2m}{n} \right\rceil\right\} + t \right] \le \exp\left\{ -t \log \frac{nt}{2m} + t - \frac{2m}{n} \right\}.$$

Proof. If $m = 0$, every degree is zero and the assertion is immediate. Assume henceforth that $m > 0$. For every integer $a \ge 0$, let A_a count the graphs in $\Omega(n, m, \boldsymbol{\ell})$ with degree a at v. When $a > \max\{\ell_v, 2m/n\}$, choose an edge vx and a vertex $w \in [n] \setminus \{v, x\}$, and replace vx by wx if wx is absent. The degree at v decreases by one, the degree at w increases by one, and all degree lower bounds remain satisfied. There are $a(n-2)$ initial choices for (vx, w), and at most $2(m-a)$ choices have $wx \in G$. Thus there are at least $na - 2m$ legal switchings from each graph with degree a at v. A reverse switching has at most $2(m - a + 1)$ choices. Double counting gives $(na - 2m)A_a \le 2(m - a + 1)A_{a-1}$, and hence $A_a \le (2m/n)A_{a-1}/(a - 2m/n)$.

Let p_q be the probability that the random variable on the left of the claimed domination equals q. For $q \ge 1$, the recurrence gives $p_q \le (2m/(nq))p_{q-1}$. When $q = 1$, use the fact that p_0 includes all degree layers up to $\max\{\ell_v, \lceil 2m/n \rceil\}$ and is therefore at least the probability of its last layer. For the Poisson probabilities π_q of mean $2m/n$, we have $\pi_q/\pi_{q-1} = 2m/(nq)$. Therefore p_q/π_q is nonincreasing. Since both probability distributions sum to one, the differences $p_q - \pi_q$ change sign at most once, from nonnegative to nonpositive. Every upper tail of (p_q) is consequently bounded by the corresponding Poisson upper tail. The final assertion follows from the Poisson upper-tail bound in Lemma 2.4. □

The next switching preserves the entire degree sequence and bounds the number of neighbors in a specified set.

Lemma 3.2. *Fix a realizable degree sequence $\boldsymbol{d} = (d_v)_{v\in[n]}$, and choose G uniformly among its simple graph realizations. Fix $B \subseteq [n]$ and $y \in [n] \setminus B$. If*

$$\sum_{v \in [n]\setminus B} d_v - 4 \max_v d_v - 2(\max_v d_v)^2 > 0,$$

then

$$|N_G(y) \cap B| \preceq \mathrm{Poisson}\left(\frac{d_y \sum_{v\in B} d_v}{\sum_{v\in[n]\setminus B} d_v - 4\max_v d_v - 2(\max_v d_v)^2} \right).$$

Proof. For each integer $t \ge 0$, let A_t count realizations of $\boldsymbol{d}$ with $|N_G(y) \cap B| = t$. For $t \ge 1$, choose $z \in N_G(y) \cap B$ and an oriented edge ab disjoint from yz, with $a \in [n] \setminus B$. Replace yz, ab by ya, zb when both new edges are absent. This preserves all degrees and decreases $|N_G(y) \cap B|$ by one.

For each of the t choices of z, there are initially $\sum_v d_v$ oriented edges ab. Exclude $\sum_{v \in B} d_v$ choices with $a \in B$, at most $4 \max_v d_v$ choices for which ab meets yz, and at most $(\max_v d_v)^2$ choices for each of $ya \in G$ and $zb \in G$. A reverse switching is specified by an edge ya and an oriented edge zb with $z \in B$. Hence

$$t \left(\sum_{v \in [n] \setminus B} d_v - 4 \max_v d_v - 2(\max_v d_v)^2 \right) A_t \le d_y \left(\sum_{v \in B} d_v \right) A_{t-1}.$$

The Poisson comparison used in Lemma 3.1 proves the assertion. □

We also need a lower bound on the probability of the degree constraints, to control the cost of conditioning on them.

Lemma 3.3. *For every fixed $A \ge 0$, if $(n, m, \boldsymbol{\ell})$ is near-critical, then a uniform m-edge graph G on $[n]$ satisfies*

$$\mathbb{P}[G \in \Omega(n, m, \boldsymbol{\ell})] \ge \exp\{-n^{1/10}\}$$

for all sufficiently large n depending only on A.

Proof. First sample each pair independently with probability $p = (1 - n^{-1/4})m / \binom{n}{2}$. The degree of each vertex then has mean $(1 + o(1)) \log n$. The lower-tail bound in Lemma 2.4 gives $\mathbb{P}_p[d_G(v) < \ell_v] \le n^{-\varphi(0.01)+o(1)}$. Harris's inequality applies to the increasing events $\{d_G(v) \ge \ell_v\}$. Since $\log(1 - x) \ge -2x$ for $0 \le x \le 1/2$, it follows that

$$\begin{aligned} \mathbb{P}_p[d_G(v) \ge \ell_v \text{ for every } v \in [n]] &\ge \prod_{v \in [n]} \mathbb{P}_p[d_G(v) \ge \ell_v] \\ &\ge \left(1 - n^{-\varphi(0.01)+o(1)}\right)^n \ge \exp\{-n^{1-\varphi(0.01)+o(1)}\}. \end{aligned}$$

The number of sampled edges has mean $(1 - n^{-1/4})m$. Since $m = (1 + \frac{n^{-1/4}}{1-n^{-1/4}})\mathbb{E}_p[|G|]$, Lemma 2.4(iii) gives $\mathbb{P}_p[|G| > m] \le \exp\{-\Omega(n^{1/2} \log n)\}$. The degree-constraint event is increasing, and therefore its probability in the uniform fixed-size model is nondecreasing in the number of edges. Consequently,

$$\mathbb{P}_p[d_G(v) \ge \ell_v \text{ for every } v] \le \mathbb{P}[G \in \Omega(n, m, \boldsymbol{\ell})] + \mathbb{P}_p[|G| > m].$$

Since $1 - \varphi(0.01) < 1/10$, subtracting the second term proves the lemma. □

These estimates give the degree bounds needed after deleting a bounded number of vertices.

Proposition 3.4. *Fix $A \ge 0$, and let $(n, m, \boldsymbol{\ell})$ be admissible and near-critical. Choose F_* uniformly from $\Omega(n, m, \boldsymbol{\ell})$, and put $B_* := \{v \in [n] : d_{F_*}(v) \le 0.03 \log n\}$. There is an absolute constant C such that, with probability $1 - o(1)$, the following statements hold.*

(i) We have $\Delta(F_*) \le C \log n$ and $|B_*| \le n^{1/4}$.

(ii) For every $y \in [n]$, we have $|N_{F_*}(y) \cap B_*| \le 1$.

For every fixed integer $h \ge 0$, these statements imply, for all sufficiently large n and every $Z \subseteq [n]$ with $|Z| \le h$,

$$\sum_{v \in [n] \setminus Z} \max\{0, \ell_v - d_{F_* - Z}(v)\} \le |Z|. \tag{3.1}$$

The convergence and the threshold on n depend only on A, h.

Proof. Near-criticality gives $2m/n \sim \log n$ and $\ell_v < 2m/n$ for sufficiently large n. Choose C sufficiently large that Lemma 3.1 gives $\mathbb{P}[d_{F_*}(v) > C \log n] \le n^{-3}$ for every v. A union bound then gives $\mathbb{P}[\Delta(F_*) > C \log n] \le n^{-2}$.

For the bound on $|B_*|$, sample pairs independently with probability $p = m/\binom{n}{2}$. Fix $S \subseteq [n]$ of size $t = \lceil n^{1/4} \rceil$. The random variables $|N_G(v) \setminus S|$, for $v \in S$, count disjoint sets of potential edges and are independent, with mean $(1 + o(1)) \log n$. Thus

$$\mathbb{P}_p[d_G(v) \le 0.03 \log n \text{ for every } v \in S] \le n^{-(\varphi(0.03)+o(1))t}.$$

Since $\varphi(0.03) > 3/4$ and $\binom{n}{t} \le (en/t)^t$, summing over S gives $\mathbb{P}_p[|\{v\colon d_G(v) \le 0.03 \log n\}| \ge t] \le \exp\{-\Omega(n^{1/4} \log n)\}$. Stirling's formula gives $\mathbb{P}_p[|G| = m] = \Theta(m^{-1/2})$. Conditioning on m edges and on the degree constraints therefore bounds the corresponding failure probability for F_* by

$$O(\sqrt{m}) \exp\{n^{1/10} - \Omega(n^{1/4} \log n)\} = o(1),$$

using Lemma 3.3.

Now condition on any degree sequence for which $\Delta(F_*) \le C \log n$ and $|B_*| \le n^{1/4}$. Given this sequence, F_* is uniform among its simple graph realizations, and B_* is fixed. In Lemma 3.2, take $B = B_* \setminus \{y\}$. The denominator is $2m - O(n^{1/4} \log n + \log^2 n) = \Theta(n \log n)$, and the numerator is $O(n^{1/4} \log^2 n)$. The resulting Poisson mean is $O(n^{-3/4} \log n)$. A Poisson variable of mean λ has probability at most $\lambda^2/2$ of being at least two, so

$$\mathbb{P}[\exists y \in [n]\colon |N_{F_*}(y) \cap B_*| > 1 \mid \boldsymbol{d}] \le n\, O((n^{-3/4} \log n)^2) = o(1).$$

This proves (i)–(ii).

Finally, deleting at most h vertices removes at most h edges incident to any remaining vertex. A vertex outside B_* has degree greater than $0.03 \log n$, whereas its required degree is at most $0.01 \log n + A$; it still satisfies that lower bound for sufficiently large n. At the other vertices, the degree loss bounds the amount by which the lower bound fails. Therefore

$$\sum_{v \in [n] \setminus Z} \max\{0, \ell_v - d_{F_* - Z}(v)\} \le \sum_{v \in B_* \setminus Z} |N_{F_*}(v) \cap Z| = \sum_{y \in Z} |N_{F_*}(y) \cap (B_* \setminus Z)| \le |Z|.$$

This proves (3.1). □

4 Reduction to cycles through prescribed pairs

We first state the counting estimate for the graph left after the low-degree vertices have been removed. Fix an offset bound $A \ge 0$ and an admissible near-critical tuple $(n_*, m_*, \boldsymbol{\ell})$ as in Section 3. Before sampling the graph, fix a matching M_0 of s pairs on $[n_*]$, where $1 \le s \le n_*^{1/10}$. Define the set of M_0-*compatible edges* by

$$\mathcal{K}_{M_0} := \left\{ e \in \binom{[n_*]}{2} \colon |e \cap V(M_0)| \le 1 \right\}.$$

For this fixed matching, write $\Phi_{M_0}(H) := \Phi(H \cap \mathcal{K}_{M_0}, M_0)$. Each counted cycle uses $n_* - s$ ordinary edges. In every auxiliary family, compatibility continues to refer to $V(M_0)$, even when the prescribed matching changes.

Choose F_* uniformly from $\Omega(n_*, m_*, \boldsymbol{\ell})$.

Theorem 4.1. *For every fixed $A \ge 0$, there is a constant $C_A > 0$ such that, for every admissible near-critical $(n_*, m_*, \boldsymbol{\ell})$ with offset bound A and every matching M_0 chosen above, with probability $1 - o(1)$,*

$$\log \Phi_{M_0}(F_*) \ge (n_* - s) \log \frac{2m_*}{n_*} - (n_* - s) - \frac{C_A n_*}{\log n_*}.$$

The failure probability tends to zero uniformly over the tuples and matchings in the statement.

We prove this estimate in Section 9. We first use it to reduce the hitting-time theorem to counting cycles in the degree-conditioned graph. For the rest of this section, probabilities refer to the original process (G_t) on $[n]$.

For an m-edge graph F on $[n]$ with $\delta(F) \ge 2$, let $b(F)$ be the number of edges of F incident to at least one vertex of degree 2.

Lemma 4.2. *For every such graph F, $\mathbb{P}[G_{\tau_2} = F] = b(F)/(m\binom{\binom{n}{2}}{m})$.*

Proof. Indeed, conditional on $G_m = F$, the edges of F appear in a uniformly random order, so the last added edge is uniform in F. The event $\tau_2 = m$ occurs precisely when deleting that edge leaves a vertex of degree less than 2. Since $\delta(F) \geq 2$, this happens precisely when the edge is incident to a degree-2 vertex of F. Thus $\mathbb{P}[\tau_2 = m \mid G_m = F] = b(F)/m$. Multiplying by $\mathbb{P}[G_m = F] = \binom{\binom{n}{2}}{m}^{-1}$ proves the formula. □

Put $d_0 \coloneqq \lfloor 0.01 \log n \rfloor$. At time τ_2, let $B \coloneqq \{v \in [n] \colon d_{G_{\tau_2}}(v) \leq d_0\}$ and let $\mathcal{T} \coloneqq \{e \in G_{\tau_2} \colon e \cap B \neq \varnothing\}$. We reveal τ_2, the set B, and all edges in $\mathcal{T}$. The edges with both vertices outside B remain unrevealed.

The following separation lets us choose vertex-disjoint paths through the low-degree vertices.

Lemma 4.3. *With probability $1 - o(1)$, the following statements hold.*

(i) $1 \leq |B| \leq n^{1/12}$ *and* $\Delta(G_{\tau_2}) \leq C \log n$ *for an absolute constant C.*

(ii) No two vertices of B are joined by a path of length at most 2. In particular, every vertex in $[n] \setminus B$ has at most one neighbor in B.

(iii) The stopping time satisfies

$$\left| \frac{2\tau_2}{n} - \log n \right| \leq 2 \log \log n + o(1).$$

Proof. Assign independent uniform arrival times in $[0, 1]$ to the edges, and let $G(p)$ contain those whose arrival times are at most p. Set $p_{\text{low}} \coloneqq \log n/(n-1)$ and $p_{\text{high}} \coloneqq (\log n + 2 \log \log n)/(n-1)$. At p_{high}, the expected number of vertices of degree at most 1 is $O(1/\log n)$. At p_{low}, the expected number of degree-1 vertices is $(1 + o(1)) \log n$.

For the second moment at p_{low}, write $a_i \coloneqq \mathbb{P}[\text{Bin}(n-2, p_{\text{low}}) = i]$. For distinct vertices u, v, splitting according to the presence of uv gives

$$\mathbb{P}[d_{G(p_{\text{low}})}(u) = d_{G(p_{\text{low}})}(v) = 1] = (1 - p_{\text{low}})a_1^2 + p_{\text{low}} a_0^2.$$

Each marginal probability is $(1 - p_{\text{low}})a_1 + p_{\text{low}} a_0$, and $a_0/a_1 = (1 - p_{\text{low}})/((n-2)p_{\text{low}}) = O(1/\log n)$. The joint probability is therefore $(1 + o(1))$ times the product of the marginals. The first and second moment bounds imply that, with high probability,

$$G(p_{\text{low}}) \subseteq G_{\tau_2} \subseteq G(p_{\text{high}}).$$

On this event, B is contained in the set of vertices whose degree in $G(p_{\text{low}})$ is at most d_0. By Lemma 2.4(i), the probability that a given vertex belongs to this latter set is at most $n^{-\varphi(0.01)+o(1)}$, where $\varphi(0.01) > 0.9439$. Its expected size is at most $n^{1-\varphi(0.01)+o(1)} = o(n^{1/12})$. Markov's inequality gives the required bound on $|B|$.

For $t \in \{1, 2\}$, fix a labeled path of length t and condition on its edges being present in $G(p_{\text{high}})$. At each end, count incident edges in $G(p_{\text{low}})$ that are not path edges and do not join the two ends. These two sets of edge indicators are disjoint and independent of the path-edge indicators. Each count has mean $(1 + o(1)) \log n$. The probability that both counts are at most d_0 is at most $n^{-2\varphi(0.01)+o(1)}$. There are at most n^{t+1} labeled paths, so the expected number with both ends of degree at most d_0 is at most

$$n^{t+1} p_{\text{high}}^t n^{-2\varphi(0.01)+o(1)} = n^{1-2\varphi(0.01)+o(1)} (\log n)^t = o(1).$$

This proves (ii). Binomial upper tails at p_{high}, followed by a union bound over the vertices, give $\Delta(G_{\tau_2}) \leq C \log n$. The edge added at τ_2 makes at least one vertex have degree 2, so $B \neq \varnothing$ for large n.

Finally, $|G(p_{\text{low}})|$ and $|G(p_{\text{high}})|$ differ from their means by $o(n)$ with high probability. Their means are $n \log n/2$ and $n(\log n + 2 \log \log n)/2$, respectively. The sandwich proves (iii). □

For each $v \in B$, choose two distinct neighbors a_v, b_v by a fixed rule using only $\mathcal{T}$, and replace the path $a_v - v - b_v$ by the pair $a_v b_v$. On the separation event, these paths are vertex-disjoint, so $M_0 \coloneqq \{a_v b_v \colon v \in B\}$ is a matching. Delete B, put $s \coloneqq |B|$, $n_* \coloneqq n - s$, and $m_* \coloneqq \tau_2 - |\mathcal{T}|$, and identify $[n] \setminus B$ with $[n_*]$ in increasing label order. For each remaining vertex u, the degree lower bound after this deletion is

$$\ell_u \coloneqq \max\{0, d_0 + 1 - d_{\mathcal{T}}(u)\}.$$

Proposition 4.4. *Conditional on any admissible exposure $(\tau_2, B, \mathcal{T})$ for which the chosen paths are vertex-disjoint, $G_{\tau_2} - B$ is uniform in $\Omega(n_*, m_*, \boldsymbol{\ell})$. With probability $1 - o(1)$, this tuple is near-critical with offset bound 2, and M_0 is a matching with $1 \le s \le n_*^{1/10}$.*

Proof. All degree-2 vertices belong to B, and all their incident edges are recorded in $\mathcal{T}$. Thus the value of $b(F)$ in Lemma 4.2 is fixed by the exposure. Every remaining vertex lies outside B, so the requirement that B be exactly the low-degree set is equivalent to

$$d_{G_{\tau_2}-B}(u) + d_{\mathcal{T}}(u) \ge d_0 + 1 \quad \text{for every } u \in [n] \setminus B.$$

Every m_*-edge graph satisfying these inequalities completes the same exposure, and every feasible completion has the same total number of edges. By Lemma 4.2, their probabilities are equal. This proves the conditional law.

By Lemma 4.3(ii), $d_{\mathcal{T}}(u) \le 1$ for every remaining vertex. Also $n_* = n - O(n^{1/12})$, so the difference between $\lfloor 0.01 \log n \rfloor$ and $\lfloor 0.01 \log n_* \rfloor$ is at most 1 for large n. Consequently $|\ell_u - \lfloor 0.01 \log n_* \rfloor| \le 2$. By the definition of B, we have $|\mathcal{T}| \le \sum_{v \in B} d_{G_{\tau_2}}(v) \le s d_0 = O(s \log n)$, and hence

$$\frac{2m_*}{n_*} - \frac{2\tau_2}{n} = O\left(\frac{s \log n}{n}\right) = o(1).$$

Together with Lemma 4.3(iii), this proves near-criticality. The chosen paths are disjoint by (ii), and $s \le n^{1/12} \le n_*^{1/10}$ for large n. □

Proof of Theorem 1.1, assuming Theorem 4.1. Fix an admissible exposure $(\tau_2, B, \mathcal{T})$ for which the chosen paths are vertex-disjoint, the tuple $(n_*, m_*, \boldsymbol{\ell})$ is near-critical with offset bound 2, and $1 \le s \le n^{1/12}$. By Lemma 4.3 and Proposition 4.4, such exposures have total probability $1 - o(1)$. Conditional on any such exposure, $G_{\tau_2} - B$ is uniform in $\Omega(n_*, m_*, \boldsymbol{\ell})$ by Proposition 4.4. Since $s \le n_*^{1/10}$ for large n, we may apply Theorem 4.1; its uniform failure bound gives a Hamilton cycle through the fixed matching M_0 with probability $1 - o(1)$.

Replace each pair $a_v b_v$ by its chosen path $a_v - v - b_v$. These paths are disjoint, so the expansion preserves a single cycle and covers every deleted vertex once. Averaging over these exposures and adding the $o(1)$ probability of all other exposures proves that G_{τ_2} is Hamiltonian with high probability. Since $\tau_{\mathrm{HC}} \ge \tau_2$ deterministically, the two hitting times coincide with high probability. □

5 Entropy and directed edge marginals

In this section, we use entropy to bound the total deviation of the directed edge marginals of Hamilton cycles through a prescribed matching from the reciprocal of the average degree.

Theorem 5.1. *Fix constants $c > 0$, $C \ge 1$, and $C_0 \ge 1$. Let G be a simple graph on n vertices such that $d(G) \ge c \log n$ and $\Delta(G) \le C d(G)$. Let M be a matching of $1 \le s \le C_0 n^{1/10}$ prescribed pairs, and let $\mathbf{C}$ be sampled from a probability distribution on the (G, M)-respecting Hamilton cycles. Choose a root pair in M and fix an arbitrary direction on it. Suppose that, for $0 \le \xi \le 1$,*

$$\mathbb{H}[\mathbf{C}] \ge (n - s) \log d(G) - (n - s) - \xi n.$$

For every ordered pair (u, v) with $uv \in G \setminus M$, put $p_{uv} \coloneqq \mathbb{P}[\mathbf{C} \text{ uses } uv \text{ from } u \text{ to } v \text{ in its root orientation}]$. Then

$$\sum_{\substack{(u,v):\, uv \in G \\ u,v \in V(G) \setminus V(M)}} \left| p_{uv} - \frac{1}{d(G)} \right| = O_{c,C,C_0}\left(n \sqrt{\xi + n^{-9/10} + \frac{1}{\sqrt{\log d(G)}}} \right).$$

In particular, if $\xi = O((\log n)^{-1/2})$, then the right-hand side is $O(n/(\log \log n)^{1/4})$.

To prove Theorem 5.1, we encode each root-oriented cycle as a perfect matching between outgoing and incoming vertex copies. We first establish a weighted version of the random-order entropy argument of Radhakrishnan [12], then bound the error term using a coarse estimate for the conditional matching marginals.

Lemma 5.2. *Let $\mathbf{\Pi}$ be sampled from a probability distribution on the bijections between two sets of size $k \geq 1$, and put $p_{ij} \coloneqq \mathbb{P}[\mathbf{\Pi}(i) = j]$. Define $\psi(0) \coloneqq 0$, $\psi(1) \coloneqq 1$, and $\psi(a) \coloneqq -a \log a/(1-a)$ for $0 < a < 1$. Then*

$$\mathbb{H}[\mathbf{\Pi}] \leq \sum_i \mathbb{H}[\mathbf{\Pi}(i)] - k + \sum_{i,j} p_{ij}\psi(p_{ij}),$$

and $0 \leq \psi(a) \leq \sqrt{a}$ for $0 \leq a \leq 1$.

Proof. Assign independent uniform priorities $U_i \in [0,1]$ to the rows, independently of $\mathbf{\Pi}$, and reveal the images in decreasing order of priority. Let A_i be the columns still available when row i is reached. Conditional on the earlier images and all priorities, the distribution $(q_j)_j$ of $\mathbf{\Pi}(i)$ is supported on A_i. Nonnegativity of relative entropy, applied to this distribution and the probability distribution $\big(p_{ij}/\sum_{\ell \in A_i} p_{i\ell}\big)_{j \in A_i}$, gives

$$\mathbb{H}[(q_j)_j] \leq -\sum_{j \in A_i} q_j \log p_{ij} + \log \sum_{j \in A_i} p_{ij}.$$

If $p_{ij} = 0$, then $q_j = 0$ almost surely, so that term contributes zero. Averaging the first sum gives $\mathbb{H}[\mathbf{\Pi}(i)]$. Since the priorities are independent of the bijection, the chain rule therefore gives

$$\mathbb{H}[\mathbf{\Pi}] \leq \sum_i \mathbb{H}[\mathbf{\Pi}(i)] + \sum_i \mathbb{E}\left[\log \sum_{j \in A_i} p_{ij}\right]. \tag{5.1}$$

Conditional on the realized bijection and $U_i = t$, for almost every $t \in [0,1]$, the column $\mathbf{\Pi}(i)$ is available. Any other column is available precisely when the priority of its row is below t, which has probability t. Hence Jensen's inequality gives

$$\mathbb{E}\left[\log \sum_{j \in A_i} p_{ij} \,\middle|\, \mathbf{\Pi}, U_i = t\right] \leq \log\big(p_{i,\mathbf{\Pi}(i)} + (1 - p_{i,\mathbf{\Pi}(i)})t\big).$$

Integrating over t and averaging over $\mathbf{\Pi}(i)$ yields

$$\mathbb{E}\left[\log \sum_{j \in A_i} p_{ij}\right] \leq \sum_j p_{ij} \int_0^1 \log\big(p_{ij} + (1 - p_{ij})t\big)\,\mathrm{d}t = -1 + \sum_j p_{ij}\psi(p_{ij}).$$

Substitution into (5.1) proves the entropy bound. To verify $\psi(a) \leq \sqrt{a}$, write $a = x^2$. The inequality is equivalent to $1 - x^2 + 2x \log x \geq 0$ for $0 < x \leq 1$. This expression has derivative $2(1 - x + \log x) \leq 0$ and value zero at $x = 1$. The endpoint values follow by continuity. □

Fix a host graph G on n vertices, a nonempty prescribed matching M, and an arbitrary direction on a chosen root pair. A direction vector θ for the other pairs is *feasible* if some (G, M)-respecting Hamilton cycle induces these directions in its root orientation. For each feasible θ, define an auxiliary bipartite graph as follows.

(i) Replace every vertex u by an outgoing copy u^{out} and an incoming copy u^{in}.

(ii) For every prescribed pair directed from x to y, remove x^{out} and y^{in}. There remain $n - |M|$ copies of each kind.

(iii) For every edge $uv \in G \setminus M$, join u^{out} to v^{in} whenever both copies are present. Include the edge $v^{\mathrm{out}}u^{\mathrm{in}}$ whenever those copies are present as well.

Denote this fixed graph by $\widetilde{G}_\theta$, and let I_θ be its outgoing vertex class.

Now let $\mathbf{C}$ have a probability distribution on the (G, M)-respecting Hamilton cycles, and let $\mathbf{\Theta}$ record its directions on the non-root prescribed pairs. The ordinary edges of $\mathbf{C}$ define a perfect matching of $\widetilde{G}_{\mathbf{\Theta}}$, or equivalently a bijection $\mathbf{\Pi}$ between its two vertex classes. The prescribed directions and this bijection recover every edge of $\mathbf{C}$. Thus

$$\mathbb{H}[\mathbf{C}] = \mathbb{H}[\mathbf{\Theta}] + \mathbb{H}[\mathbf{\Pi} \mid \mathbf{\Theta}]. \tag{5.2}$$

The distribution of $\mathbf{\Pi}$ is induced by $\mathbf{C}$; it need not be uniform on all perfect matchings of the auxiliary graph. For each admissible θ, put $p_{ij}(\theta) \coloneqq \mathbb{P}[\mathbf{\Pi}(i) = j \mid \mathbf{\Theta} = \theta]$.

We first bound the largest conditional probabilities under a coarse entropy lower bound. This estimate will control both the correction in Lemma 5.2 and the overlap of two random cycles.

Lemma 5.3. *Fix $B \geq 0$ and $C \geq 1$. Let G be a graph on n vertices with $d(G) \to \infty$ and $\Delta(G) \leq Cd(G)$. Let M be a nonempty prescribed matching, choose a root pair in M, and fix an arbitrary direction. Let $\mathbf{C}$ have a probability distribution on the (G, M)-respecting Hamilton cycles, and define $\mathbf{\Theta}$ and $p_{ij}(\theta)$ as above. If*

$$\mathbb{H}[\mathbf{C}] \geq (n - |M|) \log d(G) - Bn,$$

then

$$\mathbb{E}\left[\sum_{i \in I_{\mathbf{\Theta}}} \max_j p_{ij}(\mathbf{\Theta})\right] = O_{B,C}\left(\frac{n}{\log d(G)}\right).$$

Proof. Write $s := |M|$. By (5.2), subadditivity, and $\mathbb{H}[\mathbf{\Theta}] \leq (s-1)\log 2$, the assumed lower bound gives

$$\mathbb{E}\left[\sum_{i \in I_{\mathbf{\Theta}}} \left(\log(Cd(G)) - \mathbb{H}[(p_{ij}(\mathbf{\Theta}))_j]\right)\right] \leq (n-s)\log C + Bn + (s-1)\log 2 = O_{B,C}(n).$$

A probability distribution on at most $Cd(G)$ points whose largest probability is a has entropy at most $\log 2 + (1-a)\log(Cd(G))$: first specify whether the most likely point occurs, and then specify the remaining outcome. Applying this bound to each conditional row, and using $\log(Cd(G)) \sim \log d(G)$, gives

$$\mathbb{E}\left[\sum_{i \in I_{\mathbf{\Theta}}} \max_j p_{ij}(\mathbf{\Theta})\right] \leq \frac{O_{B,C}(n) + (n-s)\log 2}{\log(Cd(G))} = O_{B,C}\left(\frac{n}{\log d(G)}\right). \qquad \square$$

Proof of Theorem 5.1. The entropy gap will bound the relative entropy of the conditional matching marginals. We first control the correction in Lemma 5.2, then apply Pinsker's inequality and average over the prescribed directions. For each admissible θ, put $\lambda_\theta := |\widetilde{G}_\theta|/(n-s)$.

Starting from the $2|G|$ directed edges of G, removing the arcs corresponding to $G \cap M$ and the arcs incident to the deleted vertex copies removes $O(s\Delta(G) + s)$ edges. The auxiliary graph therefore satisfies

$$\lambda_\theta = d(G)(1 + O_{C,C_0}(n^{-9/10})), \quad \text{and} \quad d_{\widetilde{G}_\theta}(i) \leq Cd(G) \tag{5.3}$$

for every outgoing copy i, uniformly over θ.

Apply Lemma 5.2 under each conditional law given $\mathbf{\Theta} = \theta$ and then average. By Lemma 5.3, with $B = 2$, its correction term is at most

$$\mathbb{E}\left[\sum_{i,j} p_{ij}(\mathbf{\Theta})\psi(p_{ij}(\mathbf{\Theta}))\right] \leq \mathbb{E}\left[\sum_i \sqrt{\max_j p_{ij}(\mathbf{\Theta})}\right] \leq \sqrt{(n-s)\,\mathbb{E}\left[\sum_i \max_j p_{ij}(\mathbf{\Theta})\right]} = O\left(\frac{n}{\sqrt{\log d(G)}}\right).$$

The first inequality uses $\psi(a) \leq \sqrt{a}$ and $\sum_j p_{ij}(\theta) = 1$. Thus

$$\mathbb{H}[\mathbf{C}] \leq \mathbb{H}[\mathbf{\Theta}] + \mathbb{E}\left[\sum_i \mathbb{H}[(p_{ij}(\mathbf{\Theta}))_j]\right] - (n-s) + O\left(\frac{n}{\sqrt{\log d(G)}}\right). \tag{5.4}$$

For each θ, the weights $p_{ij}(\theta)/(n-s)$ form a probability distribution on the edges of $\widetilde{G}_\theta$. Since $|\widetilde{G}_\theta| = (n-s)\lambda_\theta$, expanding the relative entropy gives

$$\sum_{i \in I_\theta} \mathbb{H}[\mathbf{\Pi}(i) \mid \mathbf{\Theta} = \theta] = (n-s)\log\lambda_\theta - (n-s)D_{\mathrm{KL}}\left[\left(\frac{p_{ij}(\theta)}{n-s}\right)_{ij \in \widetilde{G}_\theta} \middle\| \mathrm{Unif}(\widetilde{G}_\theta)\right].$$

Substitute this identity into (5.4), and use the assumed lower bound on $\mathbb{H}[\mathbf{C}]$. We obtain

$$\begin{aligned}
&(n-s)\mathbb{E}\left[D_{\mathrm{KL}}\left[\left(\frac{p_{ij}(\mathbf{\Theta})}{n-s}\right)_{ij \in \widetilde{G}_{\mathbf{\Theta}}} \middle\| \mathrm{Unif}(\widetilde{G}_{\mathbf{\Theta}})\right]\right] \\
&\quad \leq \xi n + \mathbb{H}[\mathbf{\Theta}] + (n-s)\mathbb{E}\left[\log\frac{\lambda_{\mathbf{\Theta}}}{d(G)}\right] + O\left(\frac{n}{\sqrt{\log d(G)}}\right) = O\left(n\left(\xi + n^{-9/10} + \frac{1}{\sqrt{\log d(G)}}\right)\right).
\end{aligned}$$

The last equality uses (5.3) and $\mathbb{H}[\boldsymbol{\Theta}] \le (s-1)\log 2$.

For each θ, multiply the difference between $p_{ij}(\theta)/(n-s)$ and $1/|\widetilde{G}_\theta|$ by $n-s$ to obtain $p_{ij}(\theta) - 1/\lambda_\theta$. Applying Lemma 2.5 conditionally and then Cauchy–Schwarz gives

$$\mathbb{E}\left[\sum_{ij\in\widetilde{G}_{\boldsymbol{\Theta}}}\left|p_{ij}(\boldsymbol{\Theta}) - \frac{1}{\lambda_{\boldsymbol{\Theta}}}\right|\right] \le (n-s)\sqrt{2\mathbb{E}\left[D_{\mathrm{KL}}\left[\left(\frac{p_{ij}(\boldsymbol{\Theta})}{n-s}\right)_{ij\in\widetilde{G}_{\boldsymbol{\Theta}}}\middle\|\,\mathrm{Unif}(\widetilde{G}_{\boldsymbol{\Theta}})\right]\right]}$$
$$= O\left(n\sqrt{\xi + n^{-9/10} + \frac{1}{\sqrt{\log d(G)}}}\right).$$

Replacing $1/\lambda_\theta$ by $1/d(G)$ in this sum adds at most $|\widetilde{G}_\theta||1/\lambda_\theta - 1/d(G)| = (n-s)|1-\lambda_\theta/d(G)| = O_{C,C_0}(n^{1/10})$. Every ordered edge uv with $u,v \in V(G)\setminus V(M)$ has the same auxiliary representation under every θ. Its directed marginal therefore satisfies $p_{uv} = \mathbb{E}[p_{u^{\mathrm{out}},v^{\mathrm{in}}}(\boldsymbol{\Theta})]$. The triangle inequality gives $|p_{uv} - 1/d(G)| \le \mathbb{E}[|p_{u^{\mathrm{out}},v^{\mathrm{in}}}(\boldsymbol{\Theta}) - 1/d(G)|]$. Summing over these ordered edges proves the first conclusion. The final assertion follows from $d(G) \ge c\log n$. □

The following consequence records the error bound used in the edge-resampling argument.

Corollary 5.4. *Fix $c>0$, $C\ge 1$, $C_0 \ge 1$, and $B \ge 0$. Let H be a graph on n vertices with $d(H) \ge c\log n$ and $\Delta(H) \le Cd(H)$. Let $G \subseteq H$ be obtained by deleting at most $C_0 n^{1/10} d(H)$ edges, and let M be a nonempty matching of at most $C_0 n^{1/10}$ prescribed pairs. Suppose a family $\mathcal{C}$ of (G,M)-respecting Hamilton cycles satisfies*

$$\log|\mathcal{C}| \ge (n-|M|)\log d(H) - (n-|M|) - \frac{Bn}{\sqrt{\log n}}.$$

Orient a uniform cycle from $\mathcal{C}$ by a fixed root direction, and let p_{uv} be its directed edge marginals. Then

$$\sum_{\substack{(u,v):\ uv\in G\\ u,v\in V(G)\setminus V(M)}}\left|p_{uv} - \frac{1}{d(H)}\right| = O_{c,C,C_0,B}\left(\frac{n}{(\log\log n)^{1/4}}\right).$$

For every $t>0$, only $O_{c,C,C_0,B}(|H|/(t(\log\log n)^{1/4}))$ of these ordered pairs satisfy $|d(H)p_{uv} - 1| > t$.

Proof. The edge-deletion bound gives $d(G) = d(H)(1 + O_{C_0}(n^{-9/10}))$. Consequently $\Delta(G) \le 2Cd(G)$ for sufficiently large n, and replacing $\log d(H)$ by $\log d(G)$ in the entropy lower bound changes it by only $O_{C_0}(n^{1/10})$. Apply Theorem 5.1 to G with $\xi = O_{C_0,B}((\log n)^{-1/2})$. Replacing $1/d(G)$ by $1/d(H)$ in the resulting sum adds at most $2|G||1/d(G) - 1/d(H)| = O_{C_0}(n^{1/10})$, proving the first assertion. Each ordered pair with relative deviation greater than t contributes more than $t/d(H)$ to the sum, and $nd(H) = 2|H|$, which proves the counting assertion. □

This corollary and the following overlap bound also apply to subfamilies defined by additional prescribed pair directions, provided that their cardinality lower bounds hold for the restricted family. Restricting to an n^{-A_0} fraction of the cycles, for fixed A_0, decreases the logarithmic lower bound by at most $A_0\log n$.

The overlap estimate below requires only a weaker entropy lower bound. It will control the variance of the number of cycles left after deleting a random set of edges.

Lemma 5.5. *Fix $B\ge 0$ and $C\ge 1$. Let G be a graph on n vertices with $d(G)\to\infty$ and $\Delta(G) \le Cd(G)$. Let $\mathcal{C}$ be a nonempty family of (G,M)-respecting Hamilton cycles for a fixed nonempty prescribed matching M. If*

$$\log|\mathcal{C}| \ge (n-|M|)\log d(G) - Bn,$$

then, for independent uniform cycles $\mathbf{C}_1, \mathbf{C}_2 \in \mathcal{C}$,

$$\mathbb{E}[|(\mathbf{C}_1\setminus M)\cap(\mathbf{C}_2\setminus M)|] = O_{B,C}\left(\frac{n}{\log d(G)}\right).$$

Proof. Fix a root direction and encode $\mathbf{C}_1$ by the auxiliary bijection above. Since $\mathbf{C}_1$ is uniform on $\mathcal{C}$, its entropy is $\log|\mathcal{C}|$, so Lemma 5.3 bounds $\mathbb{E}[\sum_i \max_j p_{ij}(\mathbf{\Theta})]$ by $O_{B,C}(n/\log d(G))$. For an ordinary edge e, put $q_e(\theta) \coloneqq \mathbb{P}[e \in \mathbf{C}_1 \setminus M \mid \mathbf{\Theta} = \theta]$. Its two possible directions give at most two auxiliary edges. Thus

$$\sum_{e\in G\setminus M} q_e(\theta)^2 \le 2\sum_{ij\in\widetilde{G}_\theta} p_{ij}(\theta)^2 \le 2\sum_{i\in I_\theta}\max_j p_{ij}(\theta).$$

By independence of $\mathbf{C}_1$ and $\mathbf{C}_2$, and Jensen's inequality,

$$\begin{aligned}\mathbb{E}[|(\mathbf{C}_1\setminus M)\cap(\mathbf{C}_2\setminus M)|] &= \sum_{e\in G\setminus M}\mathbb{P}[e\in\mathbf{C}_1]^2\\ &\le \mathbb{E}\left[\sum_{e\in G\setminus M} q_e(\mathbf{\Theta})^2\right] \le 2\mathbb{E}\left[\sum_i \max_j p_{ij}(\mathbf{\Theta})\right] = O_{B,C}\left(\frac{n}{\log d(G)}\right).\end{aligned}$$

This proves the lemma. □

6 Conditioned edge deletion

Fix the admissible near-critical tuple and prescribed matching from Section 4. We use two laws on edge-deletion sequences.

- $\mathbb{P}$ is the law of the sequence

$$\binom{[n_*]}{2} = H_{\binom{n_*}{2}} \supsetneq \cdots \supsetneq H_{m_*+1} \supsetneq H_{m_*},$$

  where each step deletes a uniformly chosen edge.
- $\mathbb{P}^*$ is the law of the same sequence conditional on $H_{m_*} \in \Omega(n_*, m_*, \boldsymbol{\ell})$, so $H_{m_*} = F_*$. Equivalently, sample F_* uniformly from this family and add the missing edges in a uniformly random order.

The probability of the conditioning event is

$$\beta \coloneqq \mathbb{P}[H_{m_*} \in \Omega(n_*, m_*, \boldsymbol{\ell})] = \frac{|\Omega(n_*, m_*, \boldsymbol{\ell})|}{\binom{\binom{n_*}{2}}{m_*}} \ge \exp\{-n_*^{1/10}\}, \tag{6.1}$$

where the inequality follows from Lemma 3.3.

Under $\mathbb{P}^*$, the pair (F_*, H_j) is uniform among all pairs satisfying $F_* \subseteq H_j \subseteq \binom{[n_*]}{2}$, $|F_*| = m_*$, $|H_j| = j$, and $F_* \in \Omega(n_*, m_*, \boldsymbol{\ell})$. Conditioning on the degree constraints changes the distribution of the next deleted edge. The following observation makes this change explicit.

Fact 6.1. *For a graph H on $[n_*]$, define*

$$Z_{m_*}(H) \coloneqq |\{F \subseteq H \colon F \in \Omega(n_*, m_*, \boldsymbol{\ell})\}|.$$

If $j > m_*$, $|H| = j$, and $Z_{m_*}(H) > 0$, then, for every $e \in H$,

$$\mathbb{P}^*[e \text{ is deleted next} \mid H_j = H] = \frac{Z_{m_*}(H-e)}{(j-m_*)Z_{m_*}(H)}.$$

Proof. Conditional on $H_j = H$, the graph F_* is uniform among the $Z_{m_*}(H)$ members of $\Omega(n_*, m_*, \boldsymbol{\ell})$ contained in H. Given F_*, the next edge is uniform in $H \setminus F_*$. There are $j - m_*$ such edges, and exactly $Z_{m_*}(H-e)$ choices of F_* avoid e. Averaging gives the stated probability. □

We will also delete a bounded set of vertices. Fix $D \subseteq [n_*]$ and $m_* \le j \le \binom{n_*}{2}$. Write

$$F_*(D) \coloneqq \{e \in F_* \colon e \cap D \neq \varnothing\}, \quad \text{and} \quad H_j(D) \coloneqq \{e \in H_j \colon e \cap D \neq \varnothing\},$$

and put $\hat{F}_* \coloneqq F_* - D$ and $\hat{H}_j \coloneqq H_j - D$. For each $v \in [n_*] \setminus D$, the degree supplied by the revealed edges of $F_*(D)$ leaves the lower bound $\ell'_v \coloneqq \max\{0, \ell_v - d_{F_*(D)}(v)\}$ for $d_{\hat{F}_*}(v)$. We reveal $F_*(D), H_j(D)$ and leave $\hat{F}_*, \hat{H}_j$ unrevealed.

Lemma 6.2. *Conditional on any $\mathbb{P}^*$-admissible values of $F_*(D), H_j(D)$, the following statements hold.*

(i) $\hat{F}_*$ is uniform among the graphs on $[n_*] \setminus D$ with $m_* - |F_*(D)|$ edges and $d_{\hat{F}_*}(v) \geq \ell'_v$ for every v.

(ii) Given $\hat{F}_*$, the graph $\hat{H}_j$ is uniform among the graphs on the same vertex set that contain $\hat{F}_*$ and have $j - |H_j(D)|$ edges.

(iii) For every $y \in [n_*] \setminus D$,

$$\sum_{v \in [n_*] \setminus (D \cup \{y\})} \max\{0, \ell'_v - d_{\hat{F}_* - y}(v)\} \leq \sum_{v \in [n_*] \setminus \{y\}} \max\{0, \ell_v - d_{F_* - y}(v)\}.$$

If F_* satisfies Proposition 3.4(i)–(ii), then the left side is at most one.

Proof. Deleting the fixed incident edge sets gives a bijection between the original nested pairs and the nested pairs described in (i)–(ii). Its inverse adds $F_*(D)$ and $H_j(D)$ to the inner and outer graphs, respectively. The constraints at vertices of D are already determined by $F_*(D)$. Every allowed inner graph has exactly

$$\binom{\binom{n_* - |D|}{2} - m_* + |F_*(D)|}{j - |H_j(D)| - m_* + |F_*(D)|}$$

outer extensions. This number depends only on the fixed edge counts, so restriction preserves the stated uniform distributions.

For (iii), fix $v \in [n_*] \setminus (D \cup \{y\})$. If $\ell_v \leq d_{F_*(D)}(v)$, then $\ell'_v = 0$ and the summand on the left is zero. Otherwise, since $v, y \notin D$, the pair vy is not among the exposed edges $F_*(D)$, and

$$\max\{0, \ell'_v - d_{\hat{F}_* - y}(v)\} = \max\{0, \ell_v - d_{F_*}(v) + \mathbf{1}_{\{vy \in F_*\}}\} = \max\{0, \ell_v - d_{F_* - y}(v)\}.$$

Summing gives the inequality. The last assertion follows from (3.1) with $Z = \{y\}$. □

The restriction argument also gives an exact reverse sampling law when the exposed edges form an arbitrary fixed part of the edge universe.

Lemma 6.3. *Let $\mathcal{E}$ be a finite edge universe, let $\mathcal{U} \subseteq \mathcal{E}$, and let $\mathfrak{F} \subseteq 2^{\mathcal{E}}$ be increasing. Suppose (S_*, H) is uniform among the pairs satisfying $S_* \subseteq H \subseteq \mathcal{E}$, $|S_*| = m_*$, $|H| = j$, and $S_* \in \mathfrak{F}$. Reveal $A_0 := S_* \setminus \mathcal{U}$ and $B_0 := H \setminus \mathcal{U}$. Conditional on any admissible values of these sets, the following statements hold.*

(i) The pair $(S, J) := (S_* \cap \mathcal{U}, H \cap \mathcal{U})$ is uniform among the pairs satisfying

$$S \subseteq J \subseteq \mathcal{U}, \quad |S| = m_* - |A_0|, \quad |J| = j - |B_0|, \quad A_0 \cup S \in \mathfrak{F}.$$

(ii) Fix $0 \leq t \leq |J|$ and choose T uniformly among the t-subsets of J. Conditional on admissible values of $J - T, S - T$, the pair $(S \cap T, T)$ is uniform among the pairs satisfying

$$S \cap T \subseteq T \subseteq \mathcal{U} \setminus (J - T), \quad |T| = t, \quad |S \cap T| = m_* - |A_0| - |S - T|, \quad A_0 \cup (S - T) \cup (S \cap T) \in \mathfrak{F}.$$

If $A_0 \cup (S - T) \in \mathfrak{F}$, then T is uniform among all t-subsets of $\mathcal{U} \setminus (J - T)$.

Proof. For (i), restriction is a bijection, with inverse $(S, J) \mapsto (A_0 \cup S, B_0 \cup J)$. Hence all the restricted pairs have equal probability.

For (ii), every forward triple (S, J, T) has the same probability because its three set sizes are fixed. Given the remainders, reconstruction is $S = (S - T) \cup (S \cap T)$ and $J = (J - T) \cup T$. These formulas give a bijection with the pairs in (ii). If $A_0 \cup (S - T)$ already belongs to the increasing family $\mathfrak{F}$, every subset of T having the required size may serve as $S \cap T$. Each possible T has exactly $\binom{t}{m_* - |A_0| - |S - T|}$ such choices. This number is independent of T, proving the uniform law. □

We next show that a small uniformly chosen set of edges usually preserves the lower bounds on the degrees of F_*. In the following lemma, probability is taken only over the uniformly chosen deleted edge set.

Lemma 6.4. *Fix a realization $F_* \in \Omega(n_*, m_*, \boldsymbol{\ell})$ satisfying Proposition 3.4(i). Let $\mathcal{U} \subseteq \binom{[n_*]}{2}$ have size $m_0 \geq 1$, let $k \asymp n_*$ with fixed comparison constants, and let $1 \leq \nu \leq \log n_*$. Choose T uniformly among the subsets of $\mathcal{U}$ of size $\lfloor \nu m_0/k \rfloor$. Then*

$$\mathbb{P}[\exists v \in [n_*] \colon d_{F_*-T}(v) < \ell_v] \leq O(\nu n_*^{-3/4} \log n_*) + \exp\{-\Omega((\log n_*)^2)\}.$$

The implicit constants may depend on the comparison constants for k. If every edge of $\mathcal{U}$ avoids a bounded set D, then the same bound holds for the probability that $\hat{F}_* - T$ violates one of the lower bounds ℓ'_v in Lemma 6.2.

Proof. Put $B_* := \{v \colon d_{F_*}(v) \leq 0.03 \log n_*\}$. The bounds in Proposition 3.4(i) show that at most $Cn_*^{1/4} \log n_*$ edges of F_* meet B_*. Each eligible edge enters T with probability $|T|/m_0 \leq \nu/k = O(\nu/n_*)$. Thus the probability that T contains any such edge is $O(\nu n_*^{-3/4} \log n_*)$. Outside this event, every vertex of B_* keeps its degree in F_*.

A vertex outside B_* has degree greater than $0.03 \log n_*$, whereas $\ell_v = 0.01 \log n_* + O(1)$. Failure therefore requires at least $a := \lfloor 0.01 \log n_* \rfloor$ incident edges of F_* to enter T, for sufficiently large n_*. Any specified a eligible edges all enter T with probability at most $(|T|/m_0)^a$. Since $\Delta(F_*) \leq C \log n_*$, the probability of failure at a fixed vertex is at most

$$\binom{\lceil C \log n_* \rceil}{a} \left(\frac{C\nu}{n_*}\right)^a = \exp\{-\Omega((\log n_*)^2)\}.$$

If $|T| < a$, this probability is zero. Summing over vertices gives the claimed probability bound. When T avoids D, the exposed edges $F_*(D)$ are unchanged. Subtracting their contribution converts the original degree constraints into the constraints ℓ'_v, proving the last assertion. □

We now estimate how many available neighbors can lie in an excluded set determined without examining the neighborhood itself. Fix $D \subseteq [n_*]$ of bounded size, $m_* \leq j \leq \binom{n_*}{2}$, and $y \in [n_*] \setminus D$. As above, write $\hat{F}_* = F_* - D$ and $\hat{H}_j = H_j - D$.

Lemma 6.5. *Fix $c > 0$ and condition on $\mathbb{P}^*$-admissible values of*

$$F_*(D), \quad H_j(D), \quad \hat{F}_* - y, \quad \hat{H}_j - y.$$

Suppose

$$\sum_{v \in [n_*] \setminus (D \cup \{y\})} \max\{0, \ell'_v - d_{\hat{F}_*-y}(v)\} \leq 1, \quad d_{\hat{H}_j}(y) \geq c \log n_*.$$

Let $\Gamma \subseteq [n_*] \setminus (D \cup \{y\})$ be determined by the four revealed edge sets and fixed labels. If $|\Gamma|/(n_* - |D| - 1) \leq \rho = o(1)$, then

$$\mathbb{P}^*\left[|N_{\hat{H}_j}(y) \cap \Gamma| \geq \frac{2}{3} d_{\hat{H}_j}(y) \,\middle|\, F_*(D), H_j(D), \hat{F}_* - y, \hat{H}_j - y\right] \leq (C\rho)^{d_{\hat{H}_j}(y)/4}.$$

The constant C and the threshold on n_* depend only on c and the fixed bound on $|D|$.

Proof. The revealed edge counts determine $b := d_{\hat{F}_*}(y)$ and $q := d_{\hat{H}_j}(y)$. By Lemma 6.2, we can sample the two neighborhoods by first choosing a uniform b-subset of $[n_*] \setminus (D \cup \{y\})$ subject to the remaining degree constraints, and then choosing a uniform q-subset containing it. The number of such extensions is the same for every b-subset. The constraint at y depends only on b and is satisfied because the revealed values have positive probability. For the other vertices there are two cases.

(i) If the displayed sum is zero, every other vertex already satisfies its lower bound. We choose the smaller neighborhood uniformly from all b-subsets, and then choose the larger neighborhood uniformly from its q-element supersets.

(ii) If the displayed sum is one, there is a unique vertex w with $d_{\hat{F}_*-y}(w) = \ell'_w - 1$. Feasibility is exactly the requirement $w \in N_{\hat{F}_*}(y)$. We therefore choose the smaller neighborhood uniformly from the b-subsets containing w, and then choose the larger neighborhood uniformly from its q-element supersets.

In the first case, $N_{\hat{H}_j}(y)$ is already a uniform q-subset. In the second case $b \geq 1$, and we couple it to such a subset as follows. Let I_0 be a uniform b-subset of $[n_*] \setminus (D \cup \{y\})$. If $w \in I_0$, let π be the identity. Otherwise choose a uniformly from I_0 and let π exchange a, w. By symmetry, $I_1 := \pi(I_0)$ is uniform among the b-subsets containing w. Given I_0 and π, sample a uniform q-subset J_0 containing I_0 and put $J_1 := \pi(J_0)$. The permutation is a bijection between the two extension spaces, so J_1 has the required conditional law of $N_{\hat{H}_j}(y)$ and differs from J_0 by at most one replacement.

For sufficiently large n_*, we have $q \geq c \log n_* \geq 6$. In the second case, $|J_1 \cap \Gamma| \geq 2q/3$ implies $|J_0 \cap \Gamma| \geq 2q/3 - 1 \geq q/2$; in the first case, no replacement is needed. Thus the event in the lemma forces a uniform q-subset to contain at least $q/2$ elements of Γ. By Lemma 2.4(ii), its probability is at most $(2e\rho)^{q/2} \leq (C\rho)^{q/4}$. If $\rho = 0$, then Γ is empty and the event is impossible. □

We finish by controlling degrees throughout the conditioned edge-deletion sequence.

Proposition 6.6. *For each fixed integer $h \geq 0$, there are constants $c_h, C_h > 0$ such that, with $\mathbb{P}^*$-probability $1 - o(1)$, the following statements hold simultaneously for every $m_* \leq j \leq \binom{n_*}{2}$.*

(i) For every $v \in [n_*]$, $2c_h j/n_* \leq d_{H_j}(v) \leq 2C_h j/n_*$.

(ii) For every $D \subseteq [n_*]$ with $|D| \leq h$ and every $v \in [n_*] \setminus D$,

$$c_h d(H_j - D) \leq d_{H_j - D}(v) \leq C_h d(H_j - D), \quad \text{and} \quad d(H_j - D) = \frac{2j}{n_*}\big(1 + O_h(n_*^{-1})\big).$$

Proof. First condition on an F_* satisfying Proposition 3.4(i). Given F_*, the set $H_j \setminus F_*$ is a uniform $(j - m_*)$-subset of $\binom{[n_*]}{2} \setminus F_*$. For a fixed vertex v, its new incidences are hypergeometric with mean

$$\mathbb{E}[d_{H_j}(v) - d_{F_*}(v) \mid F_*] = \frac{(j - m_*)(n_* - 1 - d_{F_*}(v))}{\binom{n_*}{2} - m_*} \leq \frac{4j}{n_*}.$$

Also $d_{F_*}(v) \leq C \log n_* \leq (2C + o(1))j/n_*$. Choose $C_h > \max\{4C, 4e\}$ so large that $C_h \log(C_h/(4e)) > 12$. By Lemma 2.4(ii),

$$\mathbb{P}^*\left[d_{H_j}(v) > \frac{2C_h j}{n_*} \,\middle|\, F_*\right] \leq \left(\frac{4e}{C_h}\right)^{C_h j/n_*} \leq n_*^{-5}$$

for sufficiently large n_*, since $j/n_* \geq (1 + o(1)) \log n_*/2$. Summing over the at most $n_*\binom{n_*}{2}$ vertex–time pairs gives the upper bound in (i).

For $j \leq L_0 m_*$, where L_0 is a sufficiently large fixed constant, the lower bound follows from $d_{H_j}(v) \geq d_{F_*}(v) \geq \ell_v = (0.01 + o(1)) \log n_*$. For $j > L_0 m_*$, the displayed conditional mean is at least j/n_*. Applying Lemma 2.4(i) with $a = 1/2$, and using $\varphi(1/2) \geq 1/8$, gives

$$\mathbb{P}^*\left[d_{H_j}(v) - d_{F_*}(v) < \frac{j}{2n_*} \,\middle|\, F_*\right] \leq \exp\left\{-\frac{j}{8n_*}\right\} \leq n_*^{-5},$$

after increasing L_0. The same union bound proves the lower bound in (i). Averaging over F_* is valid because Proposition 3.4(i) holds with probability $1 - o(1)$.

Finally, deleting at most h vertices removes at most h edges incident to any remaining vertex and at most $h\Delta(H_j) = O_h(j/n_*)$ edges in total. Hence

$$|H_j - D| = j\big(1 + O_h(n_*^{-1})\big), \quad \text{and} \quad d(H_j - D) = \frac{2j}{n_*}\big(1 + O_h(n_*^{-1})\big).$$

Since $j/n_* \geq (1 + o(1)) \log n_*/2$, the loss of at most h incident edges is absorbed by the degree lower bound. Enlarging C_h and decreasing c_h proves (ii). □

7 Edge resampling and candidate balance

For a fixed prescribed matching, we compare the numbers of Hamilton cycles containing an additional pair uv, with its direction specified relative to the root pair. The pair uv need not be an edge of the host graph. We show that, with high probability, the stated degree bounds and a suitable lower bound on the number of cycles imply that these numbers are nearly equal for almost all choices of u, v. To prove this,

we estimate their change after deleting a uniformly chosen set of edges and compare the forward and reverse sampling laws.

For this section and Section 8, fix the constants in Proposition 6.6 with $h = 8$. The average degree $d(H)$ uses all edges of H, including edges excluded from the counted cycles.

Definition 7.1. An *auxiliary frame* Λ consists of the following choices.

(i) A deleted set $D_\Lambda \subseteq [n_*]$ with $|D_\Lambda| \le 8$. Write $V_\Lambda := [n_*] \setminus D_\Lambda$.

(ii) A nonempty prescribed matching M_Λ on V_Λ, obtained from M_0 by removing at most two pairs and adding at most two pairs.

(iii) A root pair in M_Λ, with an arbitrary fixed direction, and, when needed, a prescribed direction on one further pair relative to that root direction.

For a graph H on V_Λ, let $\Phi_\Lambda(H)$ be the number of $(H \cap \mathcal{K}_{M_0}, M_\Lambda)$-respecting Hamilton cycles satisfying the prescribed directions. The *legal directed candidates* are

$$\mathcal{R}_\Lambda := \big\{(u \to v) \colon u, v \in V_\Lambda \setminus \big(V(M_0) \cup V(M_\Lambda)\big),\ u \ne v\big\}.$$

For $(u \to v) \in \mathcal{R}_\Lambda$, let $\Psi_\Lambda(H; u \to v)$ count the $(H \cap \mathcal{K}_{M_0}, M_\Lambda \cup \{uv\})$-respecting Hamilton cycles satisfying all directions prescribed by Λ, with uv directed from u to v in the root orientation.

The number $\Psi_\Lambda(H; u \to v)$ is defined even when $uv \notin H$. If $uv \in H$, replacing that ordinary edge by the new prescribed pair identifies the corresponding cycles with the cycles in the frame that use uv from u to v.

There are $n_*^{O(1)}$ frames. Indeed, the deleted vertices, removed pairs, added pairs, root pair, and possible second directed pair have at most $O(n_*^8)$, $O(n_*^2)$, $O(n_*^4)$, $O(n_*)$, and $O(n_*)$ choices, respectively. Also $|\mathcal{R}_\Lambda| = (1 + o(1))|V_\Lambda|(|V_\Lambda| - 1)$, because $|M_0|, |M_\Lambda| = O(n_*^{1/10})$.

At time j, write $\hat{H}_j := H_j - D_\Lambda$. Fix a constant $B \ge 4$, unchanged throughout the argument, and consider the lower bound

$$\log \Phi_\Lambda(\hat{H}_j) \ge \big(|V_\Lambda| - |M_\Lambda|\big) \log d(\hat{H}_j) - \big(|V_\Lambda| - |M_\Lambda|\big) - \frac{Bn_*}{\sqrt{\log n_*}}. \tag{7.1}$$

For $\eta > 0$, a candidate $(u \to v) \in \mathcal{R}_\Lambda$ is *η-abnormal* in a host F with $\Phi_\Lambda(F) > 0$ if

$$\left| \frac{d(F)\Psi_\Lambda(F; u \to v)}{\Phi_\Lambda(F)} - 1 \right| > \eta.$$

Otherwise, the candidate is *η-normal*.

Proposition 7.2. *Fix a frame Λ and $m_* \le j \le \binom{n_*}{2}$. Let $\mathsf{E}_{\Lambda,j}$ be the event that the following statements hold.*

(i) F_* satisfies Proposition 3.4(i)–(ii), and the degree bounds in Proposition 6.6, with $h = 8$, hold at time j for H_j and every $H_j - D$ with $D \subseteq [n_*]$ and $|D| \le 8$.

(ii) The number $\Phi_\Lambda(\hat{H}_j)$ satisfies (7.1).

(iii) More than $(\log\log\log n_*)^{-1/2}|\mathcal{R}_\Lambda|$ legal directed candidates are $(\log\log\log n_*)^{-1/2}$-abnormal in $\hat{H}_j$.

There is a constant $c > 0$, depending only on B and the constants in the degree bounds, such that

$$\mathbb{P}^*[\mathsf{E}_{\Lambda,j}] \le \exp\{-c(\log n_*)\sqrt{\log\log\log n_*}\}. \tag{7.2}$$

The same bound holds conditional on any $\mathbb{P}^*$-admissible values of $F_*(D_\Lambda), H_j(D_\Lambda)$.

We prove Proposition 7.2 using the following variance estimate for families of sets.

Lemma 7.3. *Let $\mathcal{A}$ be a nonempty finite indexed family of h-subsets of an m-element set, where $1 \le h \le m/4$. Members of $\mathcal{A}$ are counted and sampled by their indices, so distinct indices may represent the same subset. Let T be a uniform t-subset, with $0 \le t \le m/4$, and let Y_T count the indexed members of $\mathcal{A}$ avoiding T. For independent uniform indexed members A_1, A_2 of $\mathcal{A}$,*

$$\mathbb{E}[Y_T] = \frac{\binom{m-h}{t}}{\binom{m}{t}}|\mathcal{A}|, \quad \text{and} \quad \frac{\mathrm{Var}[Y_T]}{\mathbb{E}[Y_T]^2} \le \left(\exp\left\{\frac{4th}{m}\right\} - 1\right)\frac{\mathbb{E}[|A_1 \cap A_2|]}{h}.$$

Proof. A fixed member survives with probability $\binom{m-h}{t}/\binom{m}{t}$, proving the expectation formula. For a fixed pair A_1, A_2, the ratio of its joint survival probability to the square of the single-member survival probability is

$$\prod_{i=0}^{t-1} \frac{(m-2h+|A_1\cap A_2|-i)(m-i)}{(m-h-i)^2}.$$

The excess of the ith factor over one is

$$\frac{|A_1\cap A_2|(m-i)-h^2}{(m-h-i)^2} \le \frac{4|A_1\cap A_2|}{m}.$$

Thus the product is at most $\exp\{4t|A_1\cap A_2|/m\}$. Convexity on $[0,h]$ gives

$$\exp\left\{\frac{4t|A_1\cap A_2|}{m}\right\} \le 1+\left(\exp\left\{\frac{4th}{m}\right\}-1\right)\frac{|A_1\cap A_2|}{h}.$$

Averaging over the ordered pair of indexed members and subtracting one proves the variance bound. □

Proof of Proposition 7.2. If too many candidates are abnormal, deleting a uniformly chosen set of edges will usually leave a large set of abnormal missing pairs containing very few of the sampled edges. Viewed in reverse, the sampled edges are uniform in the missing-edge universe once deletion has preserved the degree lower bounds in the smaller graph. A hypergeometric lower-tail estimate will make this discrepancy unlikely.

Fix $\mathbb{P}^*$-admissible values of $F_*(D_\Lambda), H_j(D_\Lambda)$. If no realization of $\mathsf{E}_{\Lambda,j}$ has these values, then its conditional probability is zero. Otherwise Lemma 6.2 gives the following distributions.

(i) $\hat{F}_* := F_* - D_\Lambda$ is uniform among the graphs on V_Λ with $m_* - |F_*(D_\Lambda)|$ edges and the degree lower bounds ℓ'_v.

(ii) Given $\hat{F}_*$, the graph $\hat{H}_j$ is uniform among its extensions with $m := j - |H_j(D_\Lambda)|$ edges.

The values of m and $k := |V_\Lambda| - |M_\Lambda|$ are fixed by these choices. Every realization in $\mathsf{E}_{\Lambda,j}$ has $k \asymp n_*$, $m \ge cn_*\log n_*$, and $d(\hat{H}_j) = 2m/|V_\Lambda|$.

We examine the same random edge deletion in two orders. Put $\nu := \log\log\log n_*/100$ and $\varepsilon := 1/(10^5\sqrt{\log\log\log n_*})$. First sample $(\hat{F}_*, \hat{H}_j)$ under the conditional law just described, and then choose T uniformly among the t-subsets of $\hat{H}_j$, where

$$t := \left\lfloor \frac{m\log\log\log n_*}{100k} \right\rfloor.$$

Write $\widehat{\mathbb{P}}$ for this joint law. In the forward calculation we fix $(\hat{F}_*, \hat{H}_j)$ and randomize T. In the reverse calculation we fix $\hat{H}_j - T, \hat{F}_* - T$ and determine the conditional distribution of T. The required degree condition is $d_{\hat{F}_*-T}(v) \ge \ell'_v$ for every $v \in V_\Lambda$. The set T is sampled from all edges of $\hat{H}_j$, including edges excluded from the counted cycles. The reverse sampling universe is therefore the full set $\binom{V_\Lambda}{2} \setminus (\hat{H}_j - T)$.

For a remaining graph F on V_Λ, let

$$\mathcal{A}(F) := \left\{ uv \in \binom{V_\Lambda}{2} \setminus F : \text{at least one legal orientation of } uv \text{ is } \frac{1}{4\sqrt{\log\log\log n_*}}\text{-abnormal in } F \right\},$$

and put $\mathcal{A}(F) = \varnothing$ when $\Phi_\Lambda(F) = 0$. Define W to be the event that

(W1) $d_{\hat{F}_*-T}(v) \ge \ell'_v$ for every $v \in V_\Lambda$;

(W2) $|\mathcal{A}(\hat{H}_j - T)| \ge \frac{1}{8\sqrt{\log\log\log n_*}} \left|\binom{V_\Lambda}{2} \setminus (\hat{H}_j - T)\right|$;

(W3) $|T \cap \mathcal{A}(\hat{H}_j - T)| \le (\log\log\log n_*)^{-1} t$.

The forward and reverse estimates for W are established in the claims below. For the remainder of the proof, call a candidate *large* if $\Psi_\Lambda(\hat{H}_j; u \to v) \ge \Phi_\Lambda(\hat{H}_j) n_*^{-200}$.

Claim 7.4. *There is a constant $C_1 > 0$, depending only on the fixed constants in Proposition 7.2, such that the following statements hold for every realization of $\mathsf{E}_{\Lambda,j}$. Put $p_{n_*} := C_1(\log\log\log n_*)/(\log\log n_*)^{24/25}$.*

(i) With probability at least $1-p_{n_*}$ *under* $\widehat{\mathbb{P}}[\cdot \mid \hat{F}_*, \hat{H}_j]$,

$$\Phi_\Lambda(\hat{H}_j - T) = (1 \pm \varepsilon)\frac{\binom{m-k}{t}}{\binom{m}{t}}\Phi_\Lambda(\hat{H}_j). \tag{7.3}$$

(ii) For every large candidate,

$$\widehat{\mathbb{P}}\left[\left|\frac{\binom{m}{t}\Psi_\Lambda(\hat{H}_j - T; u \to v)}{\binom{m-k+1}{t}\Psi_\Lambda(\hat{H}_j; u \to v)} - 1\right| > \varepsilon \,\middle|\, \hat{F}_*, \hat{H}_j\right] \le p_{n_*}.$$

(iii) For every candidate in (ii) with $uv \in \hat{H}_j$,

$$\widehat{\mathbb{P}}\left[\left|\frac{\binom{m-1}{t-1}\Psi_\Lambda(\hat{H}_j - T; u \to v)}{\binom{m-k}{t-1}\Psi_\Lambda(\hat{H}_j; u \to v)} - 1\right| > \varepsilon \,\middle|\, \hat{F}_*, \hat{H}_j,\ uv \in T\right] \le p_{n_*}.$$

Proof. Each cycle in a large candidate family has $k-1$ ordinary edges, and

$$\log \Psi_\Lambda(\hat{H}_j; u \to v) \ge (k-1)\log d(\hat{H}_j) - O(n_*).$$

Indeed, take logarithms of its assumed size lower bound and substitute (7.1); the additional loss $200\log n_*$ is $o(n_*)$. Passing from $\hat{H}_j$ to $\hat{H}_j \cap \mathcal{K}_{M_0}$ deletes only $O(n_*^{1/10} d(\hat{H}_j))$ edges; the prescribed pairs are excluded from the ordinary edges by definition. Thus the allowed host has average degree $d(\hat{H}_j)(1+O(n_*^{-9/10}))$ and maximum degree $O(d(\hat{H}_j))$. The change in the logarithmic reference value is $O(n_*^{1/10})$. Consequently Lemma 5.5 applies to the main family and to each large candidate family, with all prescribed directions included in their actual cardinalities.

Apply Lemma 7.3 in the ambient m-edge set $\hat{H}_j$. For the main family, use the ordinary-edge sets $C \setminus M_\Lambda$, each of size $h = k$. For a candidate $(u \to v)$, use the sets $C \setminus (M_\Lambda \cup \{uv\})$ from its cycle family, each of size $h = k-1$. Since $t \le \nu m/k$, that lemma and Lemma 5.5 give relative variance

$$O\left(\frac{\exp\{4\nu\}}{\log d(\hat{H}_j)}\right) = O\left(\frac{1}{(\log\log n_*)^{24/25}}\right).$$

Here $\exp\{4\nu\} = (\log\log n_*)^{1/25}$ and $d(\hat{H}_j) \ge c\log n_*$. Also $k, t \le m/4$ for large n_*. By Lemma 7.3, the main-family mean is $\binom{m-k}{t}\Phi_\Lambda(\hat{H}_j)/\binom{m}{t}$, and the candidate mean is $\binom{m-k+1}{t}\Psi_\Lambda(\hat{H}_j; u \to v)/\binom{m}{t}$. Chebyshev's inequality at relative tolerance ε gives failure probability at most p_{n_*} for each of these counts, proving (i)–(ii).

For the conditional estimate, the edge uv is prescribed in these cycles and is therefore excluded from their ordinary-edge sets. In particular, $\Psi_\Lambda(\hat{H}_j - \{uv\}; u \to v) = \Psi_\Lambda(\hat{H}_j; u \to v)$. Given $uv \in T$, the set $T \setminus \{uv\}$ is a uniform $(t-1)$-subset of $\hat{H}_j \setminus \{uv\}$. The candidate mean in this universe is $\binom{m-k}{t-1}\Psi_\Lambda(\hat{H}_j; u \to v)/\binom{m-1}{t-1}$. The same variance calculation proves (iii). □

The expectations in Claim 7.4(ii)–(iii) differ because conditioning on $uv \in T$ leaves $t-1$ edges to sample from an $(m-1)$-edge set. A candidate with no cycles retains none after deletion. For a candidate with $\Psi_\Lambda(\hat{H}_j; u \to v) > 0$, we track the change in $d(\hat{H}_j)\Psi_\Lambda(\hat{H}_j; u \to v)/\Phi_\Lambda(\hat{H}_j)$ by first replacing the counts after deletion by their means. By Lemma 7.3 and $d(\hat{H}_j - T)/d(\hat{H}_j) = (m-t)/m$, the resulting multiplier is

$$\begin{aligned}&\frac{d(\hat{H}_j - T)}{d(\hat{H}_j)}\frac{\mathbb{E}[\Psi_\Lambda(\hat{H}_j - T; u \to v) \mid \hat{F}_*, \hat{H}_j]}{\Psi_\Lambda(\hat{H}_j; u \to v)}\frac{\Phi_\Lambda(\hat{H}_j)}{\mathbb{E}[\Phi_\Lambda(\hat{H}_j - T) \mid \hat{F}_*, \hat{H}_j]} \\ &= \frac{m-t}{m}\frac{\binom{m-k+1}{t}}{\binom{m-k}{t}} = 1 + \frac{t(k-1)}{m(m-k-t+1)}.\end{aligned} \tag{7.4}$$

For a present edge uv, use instead the candidate mean conditional on $uv \in T$, while retaining the unconditional mean of the main family. The corresponding multiplier is

$$\frac{m-t}{m}\frac{\binom{m-k}{t-1}\binom{m}{t}}{\binom{m-1}{t-1}\binom{m-k}{t}} = 1 + \frac{k-1}{m-k-t+1}. \tag{7.5}$$

Both ratios are $1 + o(1/\sqrt{\log\log\log n_*})$ because $k/m = O(1/\log n_*)$ and $t/m = O(\log\log\log n_*/n_*)$.

Claim 7.5. *Uniformly over realizations of* $\mathsf{E}_{\Lambda,j}$*, the following statements hold with probability* $1-o(1)$ *under* $\widehat{\mathbb{P}}[\cdot \mid \hat{F}_*, \hat{H}_j]$.

(i) At least $0.9|\mathcal{R}_\Lambda|/\sqrt{\log\log\log n_*}$ *candidates remain* $1/(2\sqrt{\log\log\log n_*})$*-abnormal in* $\hat{H}_j - T$.

(ii) At most $t/(\log\log\log n_*)$ *sampled edges have a legal orientation that is* $1/(4\sqrt{\log\log\log n_*})$*-abnormal in* $\hat{H}_j - T$.

Proof. First assume (7.3). If a large candidate satisfies the concentration bound in Claim 7.4(ii), then (7.4) shows that the ratio $d(\hat{H}_j)\Psi_\Lambda(\hat{H}_j; u \to v)/\Phi_\Lambda(\hat{H}_j)$ changes by a multiplicative factor in $[1 - \frac{1}{10\sqrt{\log\log\log n_*}}, 1 + \frac{1}{10\sqrt{\log\log\log n_*}}]$. Thus an $(\log\log\log n_*)^{-1/2}$-abnormal candidate remains $\frac{1}{2\sqrt{\log\log\log n_*}}$-abnormal. For a smaller candidate, monotonicity gives

$$\frac{d(\hat{H}_j - T)\Psi_\Lambda(\hat{H}_j - T; u \to v)}{\Phi_\Lambda(\hat{H}_j - T)} \le \frac{d(\hat{H}_j - T)n_*^{-200}}{(1-\varepsilon)\binom{m-k}{t}/\binom{m}{t}} = o(1).$$

Indeed, $d(\hat{H}_j - T) \le n_*$ and $\binom{m}{t}/\binom{m-k}{t} \le \exp\{2\nu\}$. Such a candidate stays abnormal on the low side. Claim 7.4(ii) bounds the expected number of large candidates failing its concentration bound by $p_{n_*}|\mathcal{R}_\Lambda|$. Markov's inequality gives

$$\widehat{\mathbb{P}}\left[\text{more than } \frac{1}{10\sqrt{\log\log\log n_*}}|\mathcal{R}_\Lambda| \text{ large candidates fail} \,\middle|\, \hat{F}_*, \hat{H}_j\right] \le 10p_{n_*}\sqrt{\log\log\log n_*} = o(1).$$

Together with (7.3), this proves the first assertion.

For the sampled edges, apply Corollary 5.4 with $H = \hat{H}_j$, $G = \hat{H}_j \cap \mathcal{K}_{M_0}$, $n = |V_\Lambda|$, and the uniform law on the cycle family counted by $\Phi_\Lambda(\hat{H}_j)$. Use $B+1$ in place of B in that corollary. Since $|V_\Lambda| = n_* - O(1)$, the error $Bn_*/\sqrt{\log n_*}$ in (7.1) is at most $(B+1)|V_\Lambda|/\sqrt{\log|V_\Lambda|}$ for large n_*. At most $O(m\sqrt{\log\log\log n_*}/(\log\log n_*)^{1/4})$ legal orientations of edges in $\hat{H}_j$ are initially $\frac{1}{100\sqrt{\log\log\log n_*}}$-abnormal. Their expected number with underlying edge in T is $O(t\sqrt{\log\log\log n_*}/(\log\log n_*)^{1/4})$. Every other legal orientation of a present edge belongs to a large family, since its count is at least $(1 - \frac{1}{100\sqrt{\log\log\log n_*}})\Phi_\Lambda(\hat{H}_j)/d(\hat{H}_j)$. Conditional on its edge entering T, Claim 7.4(iii) bounds the probability that the conditional concentration bound fails by p_{n_*}. There are at most two legal orientations per edge, so the expected number of these failures among sampled orientations is at most $2tp_{n_*}$.

If (7.3) holds, an initially $\frac{1}{100\sqrt{\log\log\log n_*}}$-normal orientation satisfying the concentration bound in Claim 7.4(iii) remains $\frac{1}{4\sqrt{\log\log\log n_*}}$-normal, by (7.5). Markov's inequality therefore bounds the probability that more than $(\log\log\log n_*)^{-1}t$ sampled edges have an abnormal legal orientation by

$$O\left(\frac{(\log\log\log n_*)^{3/2}}{(\log\log n_*)^{1/4}} + p_{n_*}\log\log\log n_*\right) = o(1).$$

The failure of (7.3) has probability $o(1)$ and is added once. This proves the sampled-edge assertion. □

Claim 7.6. *Uniformly over realizations of* $\mathsf{E}_{\Lambda,j}$ *with the fixed exposed edge sets,* $\widehat{\mathbb{P}}[\mathsf{W} \mid \hat{F}_*, \hat{H}_j] \ge 1 - o(1)$.

Proof. First suppose that (7.3) holds; by Claim 7.4(i), its conditional failure probability is $o(1)$. We check that Corollary 5.4 applies to the remaining host. The product formula for the avoidance probability gives

$$\log\frac{\binom{m-k}{t}}{\binom{m}{t}} = -\frac{kt}{m} + O\left(\frac{\nu}{d(\hat{H}_j)} + \frac{\nu^2}{n_*}\right), \quad \text{and} \quad k\log\frac{d(\hat{H}_j - T)}{d(\hat{H}_j)} = -\frac{kt}{m} + O\left(\frac{\nu^2}{n_*}\right).$$

Thus (7.3) and (7.1) give $\log\Phi_\Lambda(\hat{H}_j - T) \ge k\log d(\hat{H}_j - T) - k - (B+1)n_*/\sqrt{\log n_*}$. The maximum degree of $\hat{H}_j - T$ is at most that of $\hat{H}_j$, while $d(\hat{H}_j - T) = d(\hat{H}_j)(1 - o(1))$. Deleting edges outside $\mathcal{K}_{M_0}$ still costs $O(n_*^{1/10}d(\hat{H}_j - T))$ edges. Apply Corollary 5.4 with $H = \hat{H}_j - T$, $G = (\hat{H}_j - T) \cap \mathcal{K}_{M_0}$, and $n = |V_\Lambda|$. Since $|V_\Lambda| = n_* - O(1)$, for sufficiently large n_* the error above is at most $(B+2)|V_\Lambda|/\sqrt{\log|V_\Lambda|}$. Hence the corollary applies with error constant $B+2$.

The corollary shows that at most $O(m\sqrt{\log\log\log n_*}/(\log\log n_*)^{1/4})$ legal orientations of edges in $\hat{H}_j - T$ are $\frac{1}{4\sqrt{\log\log\log n_*}}$-abnormal. This number is $o((\log\log\log n_*)^{-1/2}n_*^2)$, since $m = O(n_*^2)$ and $(\log\log\log n_*)^{-1}(\log\log n_*)^{1/4} \to \infty$. Subtract them from the persistent abnormal candidates in Claim 7.5. Each missing pair has at most two orientations, so, with probability $1 - o(1)$,

$$|\mathcal{A}(\hat{H}_j - T)| \geq \frac{1}{8\sqrt{\log\log\log n_*}}\binom{|V_\Lambda|}{2} \geq \frac{1}{8\sqrt{\log\log\log n_*}}\left|\binom{V_\Lambda}{2} \setminus (\hat{H}_j - T)\right|.$$

This also explains why the argument applies when the host is dense: the full edge universe bounds the missing-edge universe from above. The sampled-edge assertion in Claim 7.5 gives condition (W3) in the definition of W.

Finally, Lemma 6.4, with $\mathcal{U} = \hat{H}_j$ and $\nu = \log\log\log n_*/100$, gives condition (W1) in the definition of W with probability $1 - o(1)$. A union bound over these failures proves the claim. □

Claim 7.7. *For every admissible pair of remainders $\hat{H}_j - T, \hat{F}_* - T$, $\widehat{\mathbb{P}}[\mathsf{W} \mid \hat{H}_j - T, \hat{F}_* - T] \leq \exp\{-t/(64\sqrt{\log\log\log n_*})\}$.*

Proof. If $\hat{F}_* - T$ violates one of its lower bounds ℓ'_v, or if condition (W2) fails, then W is impossible. Otherwise apply Lemma 6.3(ii) with $\mathcal{E} = \binom{[n_*]}{2}$, $\mathcal{U} = \binom{V_\Lambda}{2}$, $(S_*, H) = (F_*, H_j)$, and $\mathfrak{F}$ the increasing family of graphs satisfying the original degree lower bounds. The restricted sets (S, J) are $(\hat{F}_*, \hat{H}_j)$, and the fixed exposed sets (A_0, B_0) are $(F_*(D_\Lambda), H_j(D_\Lambda))$. The degrees on D_Λ are already fixed by A_0. Thus condition (W1) is equivalent to $A_0 \cup (S - T) \in \mathfrak{F}$, and the lemma gives

$$T \mid \hat{H}_j - T, \hat{F}_* - T \;\sim\; \mathrm{Unif}\binom{\binom{V_\Lambda}{2} \setminus (\hat{H}_j - T)}{t}.$$

The set $\mathcal{A}(\hat{H}_j - T)$ is fixed under this conditioning. By condition (W2), the conditional mean of $|T \cap \mathcal{A}(\hat{H}_j - T)|$ is at least $t/(8\sqrt{\log\log\log n_*})$. For large n_*, $(\log\log\log n_*)^{-1}t$ is at most half this mean. By Lemma 2.4(i), with $a = 1/2$ and $\varphi(1/2) \geq 1/8$, the conditional probability is at most $\exp\{-t/(64\sqrt{\log\log\log n_*})\}$, as claimed. □

We now average the two conditional estimates. For each fixed realization $(\hat{F}_*, \hat{H}_j)$ in $\mathsf{E}_{\Lambda,j}$, Claim 7.6 gives $\widehat{\mathbb{P}}[\mathsf{W} \mid \hat{F}_*, \hat{H}_j] \geq 1 - o(1)$. For each admissible pair $(\hat{H}_j - T, \hat{F}_* - T)$, Claim 7.7 bounds $\widehat{\mathbb{P}}[\mathsf{W} \mid \hat{H}_j - T, \hat{F}_* - T]$ by $\exp\{-t/(64\sqrt{\log\log\log n_*})\}$. Taking expectations gives

$$\begin{aligned}(1 - o(1))\mathbb{P}^*[\mathsf{E}_{\Lambda,j} \mid F_*(D_\Lambda), H_j(D_\Lambda)] \leq \widehat{\mathbb{P}}[\mathsf{E}_{\Lambda,j} \cap \mathsf{W}] \leq \widehat{\mathbb{P}}[\mathsf{W}] \\ = \mathbb{E}\big[\widehat{\mathbb{P}}[\mathsf{W} \mid \hat{H}_j - T, \hat{F}_* - T]\big] \leq \exp\left\{-\frac{t}{64\sqrt{\log\log\log n_*}}\right\}.\end{aligned}$$

For every exposed record compatible with $\mathsf{E}_{\Lambda,j}$, the degree upper bound gives $m = j(1 - O(n_*^{-1}))$. Since $j \geq m_* \sim n_* \log n_*/2$ and $k \asymp n_*$, we have $t \geq c(\log n_*)\log\log\log n_*$. Therefore $t/\sqrt{\log\log\log n_*} \geq c(\log n_*)\sqrt{\log\log\log n_*}$. Absorbing the factor $1 - o(1)$ proves the conditional form of (7.2). Averaging over the exposed edge sets proves the unconditional form. □

8 Cycle modifications and uniform edge marginals

In this section, we prove a uniform bound on the fraction of counted cycles containing any ordinary edge. We first define the logarithmic lower bound under which the estimate is needed.

Write $k := n_* - s$ for the number of ordinary edges in a M_0-respecting Hamilton cycle. Define the reference value for the logarithm of $\Phi_{M_0}(H_j)$ by

$$L_j := \log \Phi_{M_0}\left(\binom{[n_*]}{2}\right) - k\sum_{h=j+1}^{\binom{n_*}{2}} \frac{1}{h}. \tag{8.1}$$

The following calculation identifies its main term.

Fact 8.1. *For every* $m_* \le j \le \binom{n_*}{2}$,

$$L_j = k \log \frac{2j}{n_*} - k + O(s \log n_* + \log n_*).$$

Proof. By Lemma 2.2, deleting the edges with two M_0-saturated vertices from the complete graph preserves a $1 - o(1)$ fraction of its cycles through M_0. Applying Stirling's formula to the enumeration in Lemma 2.2 gives

$$\log \Phi_{M_0}\left(\binom{[n_*]}{2}\right) = k \log(n_* - 1) - k + O(s \log n_* + \log n_*).$$

The harmonic sum in (8.1) equals $\log\binom{n_*}{2}/j) + O(1/j)$. Since $k/j = O(1/\log n_*)$, subtracting k times this sum gives the stated expression for L_j. □

For a graph H on $[n_*]$ with $\Phi_{M_0}(H) > 0$ and an edge $e \in H$, the edge marginal $q_e(H)$ is the probability that a uniformly chosen cycle counted by $\Phi_{M_0}(H)$ uses e as an ordinary edge. Thus

$$q_e(H) := \frac{|\{C : C \text{ is counted by } \Phi_{M_0}(H),\ e \in C \setminus M_0\}|}{\Phi_{M_0}(H)}, \quad \text{and} \quad \sum_{e \in H} q_e(H) = k.$$

Edges outside $\mathcal{K}_{M_0}$ have marginal zero. We will bound all marginals whenever the event

$$\mathsf{A}_j := \left\{ \log \Phi_{M_0}(H_j) \ge L_j - \frac{n_*}{\sqrt{\log n_*}} \right\}$$

holds. The candidate estimates and the conditional neighbor-sampling estimates were proved without assuming A_j.

Proposition 8.2. *There is an absolute constant* $C > 0$ *such that*

$$\mathbb{P}^*\left[\exists j \in [m_*, \binom{n_*}{2}] : \mathsf{A}_j \text{ holds and } \max_{e \in H_j} q_e(H_j) > \frac{Ck}{j}\right] = o(1).$$

We prove Proposition 8.2 by modifying cycles through an added prescribed pair. Changing one endpoint preserves a fixed positive fraction of the cycles for almost every new endpoint. For a pair with two M_0-unsaturated vertices, we apply this modification twice and compare with the upper bound for typical candidates. When one vertex is M_0-saturated, we apply the modification once after contraction and then sum over its neighborhood.

Every ordinary edge of a cycle counted by $\Phi_{M_0}(H)$ has either zero or one M_0-saturated vertex. In the first case, specifying the edge as an additional prescribed pair gives the family to be compared. In the second, if $y_0 z_0 \in M_0$ and the edge is $y_0 u$, contracting $u y_0 z_0$ gives the additional prescribed pair $u z_0$ on the graph with y_0 deleted. These observations give the following two cases.

Fix a graph H on $[n_*]$ and the original prescribed matching M_0. Assume $n_* - |M_0| \ge 7$, as holds for all sufficiently large n_*. We use the following choices of deleted vertices and prescribed matching.

(C1) Put $D_0 = \varnothing$ and $M = M_0$, and choose distinct $y, z \in [n_*] \setminus V(M_0)$.

(C2) Fix $y_0 z_0 \in M_0$, put $D_0 = \{y_0\}$ and $M = M_0 \setminus \{y_0 z_0\}$, and take $z = z_0$ and $y \in [n_*] \setminus V(M_0)$.

Both orders of an original pair are allowed in (C2). In either case, write $\hat{H} := H - D_0$ and $G := \hat{H} \cap \mathcal{K}_{M_0}$. Thus the ordinary edges used below have at most one vertex in $V(M_0)$. The family to be compared consists of the cycles counted by $\Phi(G, M \cup \{yz\})$. Its matching is nonempty, even if $M = \varnothing$ in (C2). We orient these cycles by the root direction $y \to z$ and consider targets $y' \in [n_*] \setminus (D_0 \cup V(M_0) \cup \{y, z\})$.

In each cycle of this family, follow the edge at y other than yz until first reaching a vertex outside $V(M_0)$, and call that vertex a. The path P from y to a has one of two forms.

(i) In Type I, P is the ordinary edge ya.

(ii) In Type II, P has edges $yu, uu', u'a$, where $uu' \in M$ is an original prescribed pair and $yu, u'a \in G$ are ordinary edges.

Indeed, if the first vertex reached belongs to $V(M_0)$, its next edge is its prescribed pair. The ordinary edge leaving the other vertex of that pair must reach a vertex outside $V(M_0)$, since it cannot have two M_0-saturated vertices. A cycle counted by $\Phi(G, M \cup \{yz\})$ has $n_* - |M_0| - 1 \geq 6$ ordinary edges, so this path cannot return to z or an earlier vertex before reaching a.

For each legal choice of these vertex labels, define the graph H_P and the prescribed matching M_P by

$$(H_P, M_P) \coloneqq \begin{cases} (\hat{H} - y,\ M \cup \{az\}), & \text{Type I,} \\ (\hat{H} - \{y, u, u'\},\ (M \setminus \{uu'\}) \cup \{az\}), & \text{Type II.} \end{cases}$$

A legal choice has distinct labels, $a \notin V(M_0) \cup \{y, z\}$, and a matching M_P on $V(H_P)$. Let Λ_P be the frame on $V(H_P)$ with matching M_P and root direction $a \to z$. Then $\Phi_{\Lambda_P}(H_P) = \Phi(H_P \cap \mathcal{K}_{M_0}, M_P)$. Define

$$\mathcal{P}_G(y, z) \coloneqq \{P \colon P \text{ is a labeled Type I or Type II path as above, with all ordinary edges in } G\}.$$

Among the cycles counted by $\Phi(G, M \cup \{yz\})$, consider those whose path at y is P. Contracting P together with yz into the pair az is a bijection from these cycles to the cycles counted by $\Phi(H_P \cap \mathcal{K}_{M_0}, M_P)$. The inverse expands az into the fixed path followed by yz. The path at y is uniquely determined by the cycle, so summing over P gives

$$\Phi(G, M \cup \{yz\}) = \sum_{P \in \mathcal{P}_G(y,z)} \Phi(H_P \cap \mathcal{K}_{M_0}, M_P). \tag{8.2}$$

For every labeled path, $\Phi(H_P \cap \mathcal{K}_{M_0}, M_P)$ is determined by $H - y$ and the labels. In particular, it does not include an indicator for the presence of ya or yu. This distinction will allow us to sample the neighbors of y later.

For Type I there are at most $d_{\hat{H}}(y)$ present paths. For Type II, the first edge determines u, its prescribed pair determines u', and there are at most $\Delta(\hat{H})$ choices for a. Therefore

$$|\mathcal{P}_G(y, z)| \leq d_{\hat{H}}(y)(1 + \Delta(\hat{H})) = O(n_*^2). \tag{8.3}$$

Lemma 8.3. *Fix either case above and a target $y' \in [n_*] \setminus (D_0 \cup V(M_0) \cup \{y, z\})$. For each path $P \in \mathcal{P}_G(y, z)$, use the frame Λ_P defined above. Then*

$$\Phi(G, M \cup \{y'z\}) \geq \sum_{P \in \mathcal{P}_G(y,z)} \sum_{\substack{v \in V(H_P) \setminus (V(M_P) \cup V(M_0) \cup \{y'\}) \\ yv \in G}} \Psi_{\Lambda_P}(H_P; v \to y'),$$

where a summand is omitted if $y' \notin V(H_P) \setminus (V(M_P) \cup V(M_0))$. The map producing this inequality is injective jointly over the labeled paths, the vertices v, and the input cycles.

Proof. Orient an input cycle by $a \to z$. Its prescribed direction $v \to y'$ gives two intervening directed paths, A_1 from z to v and A_2 from y' to a. The cyclic order is therefore

$$a \longrightarrow z \xrightarrow{A_1} v \longrightarrow y' \xrightarrow{A_2} a.$$

Remove the prescribed pairs az, vy', restore P from y to a, insert the ordinary edge yv, and add the prescribed pair $y'z$. The result has cyclic order

$$y' \longrightarrow z \xrightarrow{A_1} v \longrightarrow y \xrightarrow{P} a \xrightarrow{A_2^{-1}} y'.$$

Thus it is one spanning cycle, as illustrated in Figure 1. Reversing A_2 changes its direction of traversal but preserves its undirected edges. The path restores exactly the deleted vertices, all restored ordinary edges belong to G, and the output contains every pair of $M \cup \{y'z\}$.

We also check that the output has no repeated edge. In the input orientation, a and v are the predecessors of z and y', respectively. Thus the ordinary edge $y'z$ is absent, because either direction would give one of these vertices a second predecessor. Every restored ordinary edge has a vertex deleted from the input. These edges are distinct from one another by the restrictions on their labels, so the output is a simple Hamilton cycle.

To recover the input, orient the output by $y' \to z$. The ordinary edge entering y determines v. Follow the oriented output cycle from y until it first reaches a vertex outside $V(M_0)$. This recovers a and the entire Type I or Type II path P, including its possible internal pair uu'. Remove that path, the edge yv, and the pair $y'z$, delete the recovered cut vertices, and insert az, vy'. This recovers the input cycle and all summation labels, proving joint injectivity. □

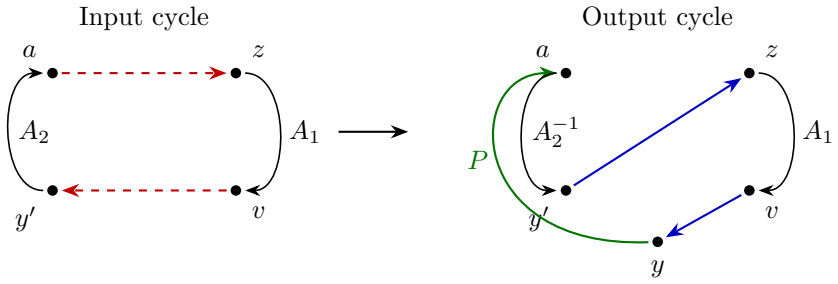


Figure 1: Replace az, vy' (dashed) by $y'z, yv$ (blue), restore P (green), and reverse A_2.

For a Type II path, constructing H_P deletes the three vertices y, u, u' from $\hat{H}$. Including D_0, every graph used in the lemma is obtained from H by deleting at most four vertices. At most two original prescribed pairs are removed and at most two new pairs are added. These choices are included in the frames of Section 7. Each application of the endpoint comparison starts with $H - D_0$ in one of Cases C1–C2; the deleted vertices of H_P are not carried into the next application.

Fix $c \in (0, 1]$ so that, whenever the conclusions of Proposition 6.6 hold, their degree bounds give $d_{H_j}(v) \geq c\, d(H_j)$ for every vertex and every time. Put $c_{\mathrm{int}} := c/24$ and $c_0 := c_{\mathrm{int}}/2$. These constants are chosen before sampling. The endpoint modification will retain a fraction at least c_{int} of its original cycle family; the estimate at a M_0-saturated vertex uses c_0.

The counting lower bound A_j also supplies the hypothesis of Proposition 7.2 for every auxiliary family that retains a sufficiently large fraction of the original cycles.

Fact 8.4. *Suppose A_j and the conclusions of Proposition 6.6, with $h = 8$, hold at time j. If a frame Λ satisfies $\Phi_\Lambda(H_j - D_\Lambda) \geq \Phi_{M_0}(H_j) n_*^{-200}$, then it satisfies (7.1).*

Proof. By Proposition 6.6(ii), $d(H_j - D_\Lambda) = d(H_j)(1 + O(1/n_*))$. Its number $|V_\Lambda| - |M_\Lambda|$ of ordinary cycle edges differs from k by $O(1)$. Hence its logarithmic reference value differs from $k \log d(H_j) - k$ by $O(\log n_*)$, and from L_j by $O(s \log n_* + \log n_*)$, using Fact 8.1. The size assumption and A_j give $\log \Phi_\Lambda(H_j - D_\Lambda) \geq L_j - n_*/\sqrt{\log n_*} - 200 \log n_*$. Its deficit from the reference value is therefore at most

$$\frac{n_*}{\sqrt{\log n_*}} + O(s \log n_* + \log n_*) + 200 \log n_* = (1 + o(1)) \frac{n_*}{\sqrt{\log n_*}}.$$

Since $B \geq 4$ was fixed before sampling, this proves (7.1) for sufficiently large n_*. □

We next define the excluded neighbors used in the two comparisons. Each rule specifies a vertex x and a set $\Gamma \subseteq [n_*] \setminus \{x\}$ determined by $H_j - x$ and fixed labels. We sample the neighborhood in the full graph H_j; all vertices removed from an auxiliary host are included in Γ.

(N1) At time j, fix one of Cases C1–C2, put $\hat{H} = H_j - D_0$, and fix vertex labels of one of the two forms of P from y to a, without requiring P to be present. Form H_P, M_P from these labels. Fix a target y'; the neighborhood to be sampled is $N_{H_j}(y)$. If $y' \notin V(H_P) \setminus (V(M_P) \cup V(M_0))$, $\Phi(H_P \cap \mathcal{K}_{M_0}, M_P) = 0$, or $d(H_P) = 0$, put $\Gamma = [n_*] \setminus \{y\}$. Otherwise, first include in Γ all vertices of $[n_*] \setminus \{y\}$ outside $V(H_P) \setminus (V(M_P) \cup V(M_0) \cup \{y'\})$. Also include every remaining vertex v satisfying

$$\Psi_{\Lambda_P}(H_P; v \to y') < \frac{\Phi(H_P \cap \mathcal{K}_{M_0}, M_P)}{2d(H_P)}.$$

(N2) Fix $y_0 z_0 \in M_0$ and $u \in [n_*] \setminus V(M_0)$. Put $G = (H_j - y_0) \cap \mathcal{K}_{M_0}$ and $M = M_0 \setminus \{y_0 z_0\}$. At y_0, let Γ consist of $V(M_0) \setminus \{y_0\}$, the vertex u, and every remaining vertex v satisfying

$$\Phi(G, M \cup \{v z_0\}) < c_0 \Phi(G, M \cup \{u z_0\}).$$

This excluded set is determined by $H_j - y_0$.

For invalid labels, set $\Gamma = [n_*] \setminus \{x\}$, where x is the sampled vertex. Use the same convention in rule (N2) if $\Phi(G, M \cup \{uz_0\}) = 0$.

These rules are fixed for all vertex labels before the graph is sampled. The next lemma establishes their conclusions simultaneously, together with the degree and candidate estimates.

Lemma 8.5. *Let* R *be the event that the following statements hold.*

(R1) The graph F_* *satisfies Proposition 3.4(i)–(ii).*

(R2) All conclusions of Proposition 6.6, with $h = 8$*, hold throughout the edge-deletion sequence.*

(R3) For every $m_* \le j \le \binom{n_*}{2}$ *and every frame satisfying (7.1), at most a* $1/\sqrt{\log\log\log n_*}$ *fraction of its legal directed candidates are* $1/\sqrt{\log\log\log n_*}$*-abnormal in* $H_j - D_\Lambda$*.*

(R4) For every time j *and every choice of labels in rules (N1)–(N2), if the resulting set* Γ *satisfies* $|\Gamma| \le (n_* - 1)/(\log\log\log n_*)^{1/16}$*, then at least* $d_{H_j}(x)/3$ *neighbors of its specified vertex* x *lie outside* Γ*.*

Then $\mathbb{P}^*[\mathsf{R}] = 1 - o(1)$*.*

Proof. By Propositions 3.4 and 6.6, the probability that either condition (R1) or condition (R2) fails is $o(1)$. Suppose both conditions hold. A failure of condition (R3) in a frame Λ at time j then implies $\mathsf{E}_{\Lambda,j}$. By Proposition 7.2, each such event has probability bounded by (7.2). There are only polynomially many choices of frames and times.

For condition (R4), there are at most $n_* + 1$ choices of D_0 and fewer than $n_*^2 + 1$ times. The Type I and Type II rules have at most n_*^4 and n_*^6 choices of their respective labels (y, z, y', a) and (y, z, y', u, u', a). The saturated-vertex rule has at most n_*^3 choices (y_0, z_0, u). Thus the total number of choices, including time, is at most

$$(n_*^2 + 1)\big((n_* + 1)(n_*^4 + n_*^6) + n_*^3\big) \le 8n_*^9.$$

For one such rule at x, reveal $F_* - x, H_j - x$. These graphs determine Γ and both degrees at x, since the full edge counts are fixed. If conditions (R1)–(R2) hold, (3.1) bounds the total unmet degree lower bound after deleting x by one, and $d_{H_j}(x) \ge c' \log n_*$ for an absolute $c' > 0$. Applying Lemma 6.5 with $D = \varnothing$ bounds the joint failure with these conditions by

$$\left(\frac{C}{(\log\log\log n_*)^{1/16}}\right)^{c' \log n_*/4} \le \exp\{-a(\log n_*)\log\log\log\log n_*\} = n_*^{-\omega(1)}$$

for an absolute $a > 0$. This uses conditioning only on the two revealed graphs, so the neighborhood retains its conditional sampling law. Summing the joint failure probabilities over frames, times, and fixed labels, and adding the probability that (R1) or (R2) fails, gives

$$\begin{aligned}\mathbb{P}^*[\mathsf{R}^c] &\le o(1) + n_*^{O(1)} \exp\{-\Omega((\log n_*)\sqrt{\log\log\log n_*})\} \\ &\quad + 8n_*^9 \exp\{-a(\log n_*)\log\log\log\log n_*\} = o(1).\end{aligned}$$

The simultaneous estimate also applies to the present paths selected later. □

Lemma 8.6. *There is an absolute constant* $C > 0$ *such that the following holds. Suppose* R *and* A_j *hold. Use the preceding construction with* $H = H_j$ *and either Case C1 or Case C2, so that* $\hat{H} = H_j - D_0$ *and* $G = (H_j - D_0) \cap \mathcal{K}_{M_0}$*. Suppose*

$$W := \Phi(G, M \cup \{yz\}) \ge \Phi_{M_0}(H_j) n_*^{-6}.$$

Then, for all but $Cn_*/(\log\log\log n_*)^{1/8}$ *vertices* $y' \in [n_*] \setminus (D_0 \cup V(M_0) \cup \{y, z\})$*,*

$$\Phi(G, M \cup \{y'z\}) \ge c_{\mathrm{int}} W.$$

Proof. We will show that, for almost every target, the path classes with sufficiently many usable neighbors contain at least half the cycles counted by W. The injection in Lemma 8.3 then gives the required comparison. Partition the family counted by W according to (8.2), and discard those present paths P for which $\Phi(H_P \cap \mathcal{K}_{M_0}, M_P) < W n_*^{-8}$. By (8.3), their total number of cycles is $O(n_*^{-6})W$. Every retained path class contains at least $\Phi_{M_0}(H_j) n_*^{-14}$ cycles, so Fact 8.4 supplies (7.1) for the frame Λ_P.

Claim 8.7. *For each retained path P, there is a set E_P of $O((\log\log\log n_*)^{-1/4}n_*)$ targets such that every target in $[n_*]\setminus E_P$ belongs to $V(H_P)\setminus(V(M_P)\cup V(M_0))$ and the excluded set Γ in rule (N1) has size at most $(\log\log\log n_*)^{-1/16}(n_*-1)$.*

Proof. First include $[n_*]\setminus\big(V(H_P)\setminus(V(M_P)\cup V(M_0))\big)$ in E_P. These are $O(s+1)=o(n_*/(\log\log\log n_*)^{1/4})$ targets. By Lemma 8.5(R3), there are at most $(\log\log\log n_*)^{-1/2}n_*^2$ abnormal directed candidates in the frame Λ_P. Summing their numbers according to their final vertex shows that at most $(\log\log\log n_*)^{-1/4}n_*$ targets have more than $(\log\log\log n_*)^{-1/4}n_*$ abnormal candidates of the form $(v\to y')$.

For any remaining target $y'\in V(H_P)\setminus(V(M_P)\cup V(M_0))$, a candidate that is not abnormal has

$$\Psi_{\Lambda_P}(H_P;v\to y')\ge\frac{(1-(\log\log\log n_*)^{-1/2})\Phi(H_P\cap\mathcal{K}_{M_0},M_P)}{d(H_P)}\ge\frac{\Phi(H_P\cap\mathcal{K}_{M_0},M_P)}{2d(H_P)}.$$

Consequently, apart from excluded labels, every excluded neighbor in (N1) gives a distinct abnormal candidate. The deleted vertices, the vertices of M_P, the original saturated vertices, and the target account for $O(s+1)$ additional labels. The excluded fraction in $[n_*]\setminus\{y\}$ is therefore

$$O((\log\log\log n_*)^{-1/4})+O\left(\frac{s+1}{n_*}\right)=o((\log\log\log n_*)^{-1/16}).$$

Adding the targets with more than $n_*/(\log\log\log n_*)^{1/4}$ abnormal candidates to E_P proves the claim. □

Claim 8.8. *For all but $O((\log\log\log n_*)^{-1/8}n_*)$ targets y' in the range of Lemma 8.6,*

$$\sum_{\substack{P\ \text{retained}\\ y'\notin E_P}}\Phi(H_P\cap\mathcal{K}_{M_0},M_P)\ge\frac{W}{2}.$$

Proof. For each retained path P, give every target in E_P weight $\Phi(H_P\cap\mathcal{K}_{M_0},M_P)$. By Claim 8.7 and (8.2),

$$\sum_{y'\in[n_*]}\sum_{\substack{P\ \text{retained}\\ y'\in E_P}}\Phi(H_P\cap\mathcal{K}_{M_0},M_P)=\sum_{P\ \text{retained}}|E_P|\Phi(H_P\cap\mathcal{K}_{M_0},M_P)\le C(\log\log\log n_*)^{-1/4}n_*W.$$

Hence at most $C(\log\log\log n_*)^{-1/8}n_*$ targets have exceptional weight greater than $(\log\log\log n_*)^{-1/8}W$. For every other target, subtracting this weight and the discarded-path contribution leaves at least $(1-(\log\log\log n_*)^{-1/8}-O(n_*^{-6}))W\ge W/2$ cycles. □

Fix a target y' for which the inequality in Claim 8.8 holds. For every retained path with $y'\notin E_P$, Lemma 8.5(R4) supplies at least $d_{H_j}(y)/3$ neighbors v outside its excluded set. Each has $yv\in G$ and $\Psi_{\Lambda_P}(H_P;v\to y')$ is at least $\Phi(H_P\cap\mathcal{K}_{M_0},M_P)/(2d(H_P))$. The degree bounds give $d_{H_j}(y)\ge c\,d(H_j)$ and $d(H_P)\le 2d(H_j)$, so these neighbors contribute at least

$$\frac{d_{H_j}(y)}{3}\frac{\Phi(H_P\cap\mathcal{K}_{M_0},M_P)}{2d(H_P)}\ge\frac{c}{12}\Phi(H_P\cap\mathcal{K}_{M_0},M_P).$$

The injection in Lemma 8.3 permits us to sum over retained paths P with $y'\notin E_P$. By Claim 8.8, these path classes contain at least $W/2$ cycles in total, yielding $\Phi(G,M\cup\{y'z\})\ge(c/24)W=c_{\mathrm{int}}W$. □

Proof of Proposition 8.2. Suppose R holds and fix $m_*\le j\le\binom{n_*}{2}$ for which A_j holds. Write $H=H_j$. We distinguish ordinary edges disjoint from $V(M_0)$ from those having exactly one M_0-saturated vertex.

Claim 8.9. *Every allowed edge $yz\in H$ disjoint from $V(M_0)$ has $q_{yz}(H)=O(1/d(H))$.*

Proof. Use Case C1, put $G=H\cap\mathcal{K}_{M_0}$, and let $W_{\max}$ be the maximum of $\Phi(G,M_0\cup\{yz\})$ over pairs outside $V(M_0)$. If $W_{\max}<\Phi_{M_0}(H)n_*^{-4}$, then $W_{\max}=o(\Phi_{M_0}(H)/d(H))$, since $d(H)=O(n_*)$. Otherwise start from a maximizing pair and apply Lemma 8.6 to change its first endpoint. The resulting family has at least $c_{\mathrm{int}}W_{\max}$ cycles. Interchange the names of its two endpoints and apply the lemma again to

change the other endpoint. Both applications are valid because every intermediate family has at least $\Phi_{M_0}(H)n_*^{-6}$ cycles for large n_*.

At the first step, at most $O(n_*/(\log\log\log n_*)^{1/8})$ endpoint choices fail, each excluding at most n_* final ordered pairs. For every other first choice, the second step has the same bound on excluded endpoints. Hence all but $O(n_*^2/(\log\log\log n_*)^{1/8})$ legal ordered pairs satisfy

$$\Phi(G, M_0 \cup \{y'z'\}) \geq c_{\mathrm{int}}^2 W_{\max}. \tag{8.4}$$

Collisions with the fixed labels or $V(M_0)$ contribute only $O((s+1)n_*) = o((\log\log\log n_*)^{-1/8}n_*^2)$ further choices.

Take the original frame Λ with $D_\Lambda = \varnothing$, $M_\Lambda = M_0$, and a fixed directed root pair. By Fact 8.4, this frame satisfies (7.1). Condition (R3) in Lemma 8.5 and Fact 2.3(i) therefore give, for all but $O((\log\log\log n_*)^{-1/2}n_*^2)$ legal ordered pairs,

$$\Phi(G, M_0 \cup \{y'z'\}) \leq \frac{2(1+(\log\log\log n_*)^{-1/2})\Phi_{M_0}(H)}{d(H)}. \tag{8.5}$$

Since $|\mathcal{R}_\Lambda| = (1-o(1))n_*^2$, some pair satisfies both (8.4) and (8.5), giving $W_{\max} = O(\Phi_{M_0}(H)/d(H))$. For an actual edge yz, designating it as an additional prescribed pair gives the exact identity $\Phi_{M_0}(H)q_{yz}(H) = \Phi(G, M_0 \cup \{yz\})$, which proves the claim. □

Claim 8.10. *Every allowed edge $y_0u \in H$ with $y_0 \in V(M_0)$ and $u \notin V(M_0)$ has $q_{y_0u}(H) = O(1/d(H))$.*

Proof. Let $y_0z_0 \in M_0$, put $G = (H - y_0) \cap \mathcal{K}_{M_0}$ and $M = M_0 \setminus \{y_0z_0\}$, and let $W := \Phi(G, M \cup \{uz_0\})$. Contracting the consecutive edges uy_0, y_0z_0 into the prescribed pair uz_0 gives $\Phi_{M_0}(H)q_{y_0u}(H) = W$. If $W < \Phi_{M_0}(H)n_*^{-4}$, the claim follows immediately. Otherwise Lemma 8.6, in Case C2 with the fixed endpoint z_0, gives $\Phi(G, M \cup \{vz_0\}) \geq c_{\mathrm{int}}W$ for all but $O((\log\log\log n_*)^{-1/8}n_*)$ vertices outside the excluded labels. Since $c_0 < c_{\mathrm{int}}$, the excluded set in rule (N2) consequently has fraction

$$O((\log\log\log n_*)^{-1/8}) + O\left(\frac{s+1}{n_*}\right) = o((\log\log\log n_*)^{-1/16}).$$

By Lemma 8.5(R4), at least $d_H(y_0)/3$ neighbors v of y_0 satisfy $\Phi(G, M \cup \{vz_0\}) \geq c_0W$.

Every counted cycle has exactly one ordinary edge at y_0. Partition the cycles by its other vertex and contract that edge together with y_0z_0. Conversely, expanding vz_0 restores the original cycle uniquely. Therefore

$$\Phi_{M_0}(H) = \sum_{v \in N_H(y_0) \setminus V(M_0)} \Phi(G, M \cup \{vz_0\}) \geq \frac{d_H(y_0)}{3} c_0 W.$$

Since $d_H(y_0) \geq c\, d(H)$, this yields $W = O(\Phi_{M_0}(H)/d(H))$ and proves the claim. □

By Claims 8.9 and 8.10, every allowed ordinary edge satisfies the required bound. Edges with two M_0-saturated vertices are forbidden and have marginal zero. Finally, $k/j = 2(1 - s/n_*)/d(H)$, so the resulting bound is $O(k/j)$. It holds simultaneously for every time satisfying A_j, because R includes all labeled rules and all times. Since $\mathbb{P}^*[\mathsf{R}] = 1 - o(1)$, this proves Proposition 8.2. □

9 Proofs of the counting theorems

We now prove Theorem 4.1. The uniform marginal bound controls the decrease in the number of cycles at each deletion. A martingale estimate bounds the accumulated decrease, including the first state at which the counting lower bound might fail.

Proof of Theorem 4.1. Recall that $k = n_* - s$, and choose $C_0 \geq 1$ at least the constant in Proposition 8.2. Define

$$\sigma^2 := C_0^2 k^2 \sum_{j=m_*+1}^{\binom{n_*}{2}} \frac{1}{j^2} = O\left(\frac{n_*}{\log n_*}\right),$$

where the estimate follows from $\sum_{j>m_*} j^{-2} \leq 1/m_*$, $k \sim n_*$, and $m_* \sim n_* \log n_*/2$.

Starting from the complete graph, stop before deleting from the first state H_j with no counted cycle. At a state with $\Phi_{M_0}(H_j) > 0$, also stop if

$$\max_{e \in H_j} q_e(H_j) > \frac{C_0 k}{j}.$$

Let j_0 be the edge count at the first stop, or m_* if no earlier stop occurs. Each deletion from a size $h > j_0$ is made from a state satisfying the marginal bound. The following estimate also includes H_{j_0}, the state reached by the last such deletion.

Claim 9.1. *For every $a > 0$,*

$$\mathbb{P}^* \left[\min_{j_0 \le j \le \binom{n_*}{2}} \{\log \Phi_{M_0}(H_j) - L_j\} < -a - \sigma^2 \right] \le \beta^{-1} \exp\left\{ -\frac{a^2}{2\sigma^2} \right\}.$$

Proof. By Fact 6.1, deletion under $\mathbb{P}^*$ is generally not uniform. We first prove the estimate under $\mathbb{P}$ and then condition on the final degree constraints. Let $\mathcal{F}_j := \sigma(H_{\binom{n_*}{2}}, \dots, H_j)$ record the deletions down to size j. As j decreases, these sigma-algebras form a filtration, meaning that $\mathcal{F}_j \subseteq \mathcal{F}_{j-1}$. Whether deletion continues from H_j is determined by $\mathcal{F}_j$. If it continues, let e_j be the uniformly deleted edge and put $D_j := q_{e_j}(H_j) - k/j$. After stopping, put all later increments equal to 0.

When deletion continues, uniformity of e_j and the identity $\sum_{e \in H_j} q_e(H_j) = k$ give

$$\mathbb{E}[q_{e_j}(H_j) \mid \mathcal{F}_j] = \frac{1}{j} \sum_{e \in H_j} q_e(H_j) = \frac{k}{j}.$$

Thus $\mathbb{E}[D_j \mid \mathcal{F}_j] = 0$ and $|D_j| \le C_0 k/j$, including after stopping. Hoeffding's exponential-moment bound [8] implies that

$$\mathbb{E}[\exp\{\lambda D_j\} \mid \mathcal{F}_j] \le \exp\left\{ \frac{\lambda^2 C_0^2 k^2}{2j^2} \right\} \quad \text{for every } \lambda > 0.$$

Let S_t be the sum of the first t increments, in decreasing order of the sizes, and let v_t be the corresponding sum of $C_0^2 k^2 / j^2$. Then $\exp\{\lambda S_t - \lambda^2 v_t/2\}$ is a nonnegative supermartingale with initial value 1. The maximal form of the Azuma–Hoeffding inequality [2], obtained by applying the supermartingale maximal inequality, gives

$$\mathbb{P}[\max_t S_t > a] \le \exp\{-\lambda a + \lambda^2 \sigma^2/2\}.$$

Taking $\lambda = a/\sigma^2$ gives $\exp\{-a^2/(2\sigma^2)\}$.

For every $h > j_0$, the exact counting identity is $\Phi_{M_0}(H_{h-1}) = \Phi_{M_0}(H_h)(1 - q_{e_h}(H_h))$. For large n_*, $C_0 k/m_* \le 1/2$, so every deletion before stopping leaves a positive number of cycles. Using $-\log(1-q) \le q + q^2$ for $0 \le q \le 1/2$, we obtain, for every $j \ge j_0$,

$$\begin{aligned} \log \Phi_{M_0}(H_j) - L_j &= \sum_{h=j+1}^{\binom{n_*}{2}} \left(\log(1 - q_{e_h}(H_h)) + \frac{k}{h} \right) \\ &\ge -\sum_{h=j+1}^{\binom{n_*}{2}} D_h - C_0^2 k^2 \sum_{h=j+1}^{\binom{n_*}{2}} \frac{1}{h^2}. \end{aligned}$$

Outside the maximal martingale event, the last expression is at least $-a - \sigma^2$. For every event E determined by the sequence,

$$\mathbb{P}^*[E] = \frac{\mathbb{P}[E \cap \{H_{m_*} \in \Omega(n_*, m_*, \ell)\}]}{\beta} \le \beta^{-1} \mathbb{P}[E].$$

This transfers the estimate to $\mathbb{P}^*$ and proves the claim. □

Take $a = n_*/\log n_*$. Since $m_* \geq n_* \log n_*/4$ for large n_* and $\sum_{j>m_*} j^{-2} \leq 1/m_*$, we have $\sigma^2 \leq 4C_0^2 n_*/\log n_*$. By (6.1), the failure probability in Claim 9.1 is therefore at most

$$\beta^{-1} \exp\left\{-\frac{a^2}{2\sigma^2}\right\} \leq \exp\left\{n_*^{1/10} - \frac{n_*}{8C_0^2 \log n_*}\right\} = o(1).$$

By Proposition 8.2, with $\mathbb{P}^*$-probability $1 - o(1)$ the marginal bound holds at every state H_j satisfying A_j. Intersect this event with the success event in Claim 9.1. Suppose A_j first fails at size j. Every preceding state H_h, with $h > j$, satisfies A_h, so its count is positive and its marginals satisfy the stopping bound. Thus no stop occurs before reaching H_j, and $j \geq j_0$. By Claim 9.1,

$$\log \Phi_{M_0}(H_j) \geq L_j - O(n_*/\log n_*),$$

contrary to $\log \Phi_{M_0}(H_j) < L_j - n_*/\sqrt{\log n_*}$. Since $\mathsf{A}_{\binom{n_*}{2}}$ holds, A_j holds throughout the sequence down to m_*. The count is therefore positive and the marginal stopping bound holds at every state. Neither stopping criterion occurs above m_*, so $j_0 = m_*$. Applying Claim 9.1 at m_* and substituting Fact 8.1, with $s \log n_* = o(n_*/\log n_*)$, proves Theorem 4.1. □

We now return to the original random graph process (G_t) on $[n]$. In the following proof, probabilities refer to that process. The lower bound in Theorem 1.2 follows by expanding the cycles through prescribed pairs; for the upper bound, we use a simultaneous first-moment estimate over its hitting-time window.

Proof of Theorem 1.2. For the upper bound, fix $n \leq m \leq \binom{n}{2}$. The complete graph contains $(n-1)!/2$ Hamilton cycles, each of which is contained in G_m with probability $\prod_{i=0}^{n-1}(m-i)/(\binom{n}{2}-i)$. Since each factor is at most $m/\binom{n}{2}$, Stirling's formula gives

$$\begin{aligned}\mathbb{E}[Z(G_m)] &= \frac{(n-1)!}{2} \prod_{i=0}^{n-1} \frac{m-i}{\binom{n}{2}-i} \\ &\leq \exp\left\{n \log \frac{2m}{n} - n + O(\log n)\right\}.\end{aligned}$$

Apply this estimate at every deterministic time in $\mathcal{I}_n := [n \log n/4, n \log n] \cap \mathbb{Z}$. Markov's inequality and a union bound give

$$\mathbb{P}\left[\exists m \in \mathcal{I}_n \colon Z(G_m) > \exp\left\{n \log \frac{2m}{n} - n + \frac{2n}{\log n}\right\}\right] \leq |\mathcal{I}_n| \exp\{-n/\log n\} = o(1).$$

By Lemma 4.3(iii), $\tau_2 \in \mathcal{I}_n$ with high probability. The simultaneous estimate therefore gives the upper bound in (1.1) at τ_2.

For the lower bound, fix an admissible exposure $(\tau_2, B, \mathcal{T})$ for which the chosen paths are vertex-disjoint, the tuple $(n_*, m_*, \boldsymbol{\ell})$ is near-critical with offset bound 2, and $1 \leq s \leq n^{1/12}$. These conditions are determined by the exposure and hold with probability $1 - o(1)$ by Lemma 4.3 and Proposition 4.4. Expanding the prescribed pairs is injective, because contracting the fixed paths $a_v - v - b_v$ recovers the core cycle. Thus Theorem 4.1, applied to the conditional law in Proposition 4.4, gives

$$\log Z(G_{\tau_2}) \geq (n_* - s) \log \frac{2m_*}{n_*} - (n_* - s) - O\left(\frac{n_*}{\log n_*}\right)$$

with failure probability tending to zero uniformly over these exposures. Here $n_* = n - s$ and $m_* = \tau_2 - |\mathcal{T}|$. Every edge of $\mathcal{T}$ meets B, whose vertices have degree at most d_0, so $|\mathcal{T}| \leq s d_0 = O(s \log n)$. Near-criticality gives $\tau_2 \sim n \log n/2$, and hence

$$\frac{n m_*}{n_* \tau_2} = \frac{1 - |\mathcal{T}|/\tau_2}{1 - s/n} = 1 + O(s/n).$$

Consequently,

$$\begin{aligned}(n_* - s) \log \frac{2m_*}{n_*} - (n_* - s) &= (n - 2s) \log \frac{2\tau_2}{n} - (n - 2s) + O(s) \\ &= n \log \frac{2\tau_2}{n} - n + O(s \log \log n) \\ &= n \log \frac{2\tau_2}{n} - n + o(n/\log n).\end{aligned}$$

Also $n_*/\log n_* = (1+o(1))n/\log n$. Averaging the conditional lower bound over these exposures and adding the $o(1)$ probability of all other exposures proves (1.1). Finally, Lemma 4.3(iii) gives

$$n\log\frac{2\tau_2}{n} - n = n\log\log n - n + O\left(\frac{n\log\log n}{\log n}\right),$$

which yields $Z(G_{\tau_2}) = (\log n/e)^n \exp\{o(n)\}$ with high probability. □

Declaration on the use of AI

GPT 6 Pro was used to assist in discussing proof strategies, checking proofs, and improving exposition.

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